

Skin Cancer Classification Using ResNet18, DCGAN-Based Data Augmentation and Explainable AI

Abstract

Medical datasets often suffer from **severe class imbalance**. When training deep learning models on these datasets, the network quickly learns to predict only the common diseases and completely **ignores the rare ones**. Unfortunately, standard accuracy metrics often hide this failure. In this study, we explore how to force a model to recognize an extreme minority class—Dermatofibroma, which has a 70:1 imbalance ratio in the HAM10000 dataset—without destroying its overall performance.

To prevent the model from **cheating by memorizing images**, we first split the dataset carefully so that multiple photos of the same physical skin lesion never crossed between the training and test sets. We then trained four ResNet18 models to track the problem. A standard baseline model achieved 77.63% overall accuracy but completely failed to find the rare class, scoring 0.0000 in recall. Next, we applied heavily weighted loss and classical image augmentations like rotations. This forced the model to pay attention, raising minority recall to 0.7000. However, because it still lacked diverse visual examples of the rare disease, the network simply started guessing the rare class everywhere. This caused precision to collapse to 0.2692 and dragged global accuracy down to 68.44%.

To provide the missing visual diversity, we trained a Deep Convolutional Generative Adversarial Network (DCGAN) to create 400 synthetic images of the rare class. When we added these synthetic images to the training set, the model stopped blindly guessing. Overall accuracy recovered to 0.7723, and the system achieved its highest overall balance with a Macro F1-score of 0.6166.

Finally, we used Grad-CAM heatmaps to look inside the network. The visual explanations confirmed that the model was anchoring its predictions on the actual shapes and borders of the skin lesions, rather than relying on background artifacts. The results demonstrate that while traditional techniques often trade precision for recall, adding synthetic morphological diversity can successfully restore balance in imbalanced medical image classification.

Chapter 1: Introduction

Skin cancer is one of the most common forms of cancer in the world. Like many cancers, the key to survival is **early detection**. When **caught early, most skin**

cancers can be treated effectively. However, if a **dangerous lesion like melanoma** is allowed to grow and spread, it becomes incredibly difficult to treat and often becomes fatal.

In standard medical practice, **dermatologists inspect the skin visually.** They use **a specialized tool called a dermatoscope**, which acts as a powerful magnifying glass with specialized lighting. This tool **allows doctors to see patterns and structures under the skin's surface that are invisible to the naked eye.** While dermoscopy is highly effective, it relies **heavily on the experience and skill of the human doctor.** Expert **dermatologists are in short supply, manual screening is slow, and diagnostic mistakes can happen when doctors are fatigued or faced with an unusual presentation of a disease.**

Because **skin cancer diagnosis is a visual task, it is a natural fit for artificial intelligence.** Over the past decade, deep learning—specifically **Convolutional Neural Networks (CNNs)—has proven to be exceptionally good at finding patterns in images.** An automated diagnostic system could act as a tireless second opinion for a doctor. It could quickly scan thousands of images, flag suspicious lesions, and help prioritize patients who need immediate attention. **The potential to save lives and reduce the workload on medical systems has made automated skin cancer classification a highly active area of research.**

However, **training a deep learning model to diagnose skin diseases involves a massive obstacle: the data itself.**

In the real world, diseases do not occur at equal rates. Common conditions, like ordinary moles, are seen in clinics every single day. Rare diseases are, by definition, seen very infrequently. When researchers collect medical images to train an artificial intelligence model, the dataset naturally reflects this real-world distribution. The result is a phenomenon known as class imbalance.

Class imbalance means that a dataset contains thousands of examples for one disease, but perhaps only a few dozen examples for another. In standard machine learning, this is a catastrophic problem. Neural networks learn by attempting to minimize their overall error. If a dataset consists of 98 ordinary moles and 2 rare cancers, the network quickly realizes something dangerous: it can achieve 98% accuracy by simply guessing "ordinary mole" every single time it sees a new image.

The network takes the path of least mathematical resistance. It learns the features of the majority class perfectly but remains completely blind to the minority class.

This leads to a secondary problem: the illusion of accuracy. **In most general machine learning projects, overall accuracy is the main metric used to decide if a model is good. But in an imbalanced medical dataset, a model with 98% accuracy is entirely useless if the 2% it gets wrong are the fatal cancers. In healthcare, missing a rare disease (a false negative) is far worse than accidentally flagging a healthy mole for review (a false positive).** Therefore, automated diagnostic systems cannot be evaluated by accuracy alone. They must be evaluated on how well they balance their ability to detect both common and rare conditions.

To study this problem, this project focuses on the HAM10000 dataset. HAM10000 stands for "**Human Against Machine with 10000 training images.**" **It is a large, public collection of dermoscopic images containing seven different types of skin lesions.**

Within this dataset, the **class imbalance is extreme.** The most **common class, Nevus (ordinary moles), contains over 6,700 images.** On the other end of the spectrum is a **class called Dermatofibroma (a benign but distinct type of skin bump).** **The dataset contains only about 70 images of Dermatofibroma.**

This creates a staggering **70:1 imbalance ratio.** **If we want to build a reliable diagnostic tool, we have to figure out how to teach a neural network to recognize a visual pattern from just 70 images, while it is simultaneously being flooded with thousands of images of other diseases.**

This project was necessary because simply feeding raw medical data into a deep learning model does not work in the real world. We need to understand the mechanics of **why these models fail on rare classes and, more importantly, how we can fix them.** We need to know if mathematical tricks are enough, or if **we need to artificially create new data to teach the model effectively.**

To answer these questions, this project was designed as an intellectual journey. Rather than just building one model and reporting the final numbers, we built a sequence of experiments. Each experiment attempts to fix the failures of the previous one.

We begin our journey by **training a standard ResNet18 model on the raw data.** **ResNet18 is a highly capable image recognition network,** and it serves as our baseline. We expect this **baseline model to fail at detecting the rare class,** but we need to observe exactly how and why it fails.

Once we understand the baseline failure, we attempt to fix the problem using a mathematical intervention called Weighted Loss. **In this second experiment, we change the rules of the training process. We heavily penalize the network for**

making mistakes on the rare class, forcing it to pay attention.

Next, we explore whether we can improve the situation by physically altering the images we have. **In the third experiment, we combine the weighted penalties with classical data augmentation.** This involves **taking our rare images and rotating, flipping, and zooming them to create slightly different versions.** This is a standard industry practice, but we want to see if it is truly enough to teach the network the boundaries of a rare disease.

When classical augmentation inevitably shows its limits—because rotating an image does not actually create new biological information—we turn to Generative Artificial Intelligence. In our fourth major phase, we train a Deep Convolutional Generative Adversarial Network (DCGAN). Instead of just twisting existing images, a GAN learns the underlying visual rules of a disease and **hallucinates entirely new, synthetic images.** We use this GAN to generate **hundreds of new examples of the rare Dermatofibroma class.** We then train a final Hybrid model on a mix of real and synthetic images to see if providing artificial visual diversity restores balance to the system.

Finally, **even if a model achieves high scores on paper, doctors cannot trust a "black box" that gives no explanation for its decisions. An AI might correctly classify an image not because it recognized the tumor, but because it noticed a surgical ruler or a dark shadow in the background of the photograph. To ensure our model is making decisions based on actual medical features, we end the project by looking inside the network using Explainable AI. Specifically, we use a technique called Grad-CAM to generate visual heatmaps. These heatmaps show us exactly which pixels the neural network was looking at when it made its diagnosis.**

The core research objective of this project is to determine the most effective way to recover a network's ability to detect an extreme minority class without destroying its ability to classify the majority classes.

Our hypothesis is this: Mathematical penalties and classical image rotations are not enough to solve extreme class imbalance because they do not provide new structural information. By generating synthetic morphological diversity using a DCGAN, we can teach the network the true visual boundaries of a rare disease, thereby restoring the overall balance and reliability of the medical diagnostic system.

Through this sequence of experiments, we aim to uncover the practical realities, tradeoffs, and limits of training deep learning models on imbalanced medical data.

Research Contributions

This project contributes to the field of medical image classification through the following key achievements:

- **Strict Data Leakage Prevention:** Implemented a lesion-level splitting strategy that maps unique physical lesion identifiers, ensuring that overlapping images of the same skin condition do not cross between training and testing sets to artificially inflate performance.
- **Ablation of Imbalance Techniques:** Conducted a step-by-step comparative study demonstrating exactly how and why traditional class-imbalance solutions—such as loss weighting and classical data augmentation—cause a neural network to trade precision for recall.
- **Generative Hybrid Training:** Successfully trained a DCGAN on an extremely limited micro-dataset of 71 images by applying stability interventions, and demonstrated that injecting synthetic morphological diversity improves the overall balanced performance (Macro F1) of a diagnostic system.
- **Visual Interpretability Verification:** Applied Grad-CAM to validate the spatial attention of the final convolutional layers, confirming that the trained model anchors its predictions on organic lesion boundaries rather than background photographic artifacts.

Chapter 3: Dataset Description and Data Integrity Framework

Before we can build an artificial intelligence model, we must thoroughly understand the data we are feeding into it. **In machine learning, an algorithm is only as good as the data it learns from. If the data is biased, the model will be biased. If the data contains hidden shortcuts, the model will cheat.**

In this chapter, we explore the HAM10000 dataset. We will look at the different types of skin diseases it contains, explain why the dataset is so severely imbalanced, and highlight why we selected the rarest disease as the ultimate test for our project. **Most importantly, we will expose a hidden trap in medical datasets called "data leakage." We will explain exactly why standard data-splitting methods fail in healthcare, and how we built a strict, lesion-level splitting framework to ensure our experimental results are completely trustworthy.**

The HAM10000 Dataset

The HAM10000 dataset (Human Against Machine with 10000 training images) is one

of the most famous public collections of dermoscopic skin images. **It was created to help researchers train computer vision systems to diagnose skin cancer. The dataset contains slightly over 10,000 high-resolution color photographs of skin lesions, taken using a specialized magnifying tool called a dermatoscope.**

Every image in the dataset is categorized into one of seven different classes.

To understand what the neural network is trying to learn, it is helpful to understand what these seven diseases are in plain English:

1. **Melanocytic Nevi (nv):** These are **ordinary, harmless moles**. Almost everyone has them. Because they are **so common in everyday life, they make up the vast majority of the dataset**.
2. **Melanoma (mel):** This is the **deadliest form of skin cancer**. It can **spread quickly to other organs if not caught early**. Visually, it often looks like **an irregular, dark, and asymmetrical mole**.
3. **Basal Cell Carcinoma (bcc):** This is a **very common type of skin cancer**. It usually **appears as a slightly transparent or shiny bump on sun-exposed skin**. **While highly treatable, it must be surgically removed**.
4. **Actinic Keratoses and Intraepithelial Carcinoma (akiec):** These are rough, scaly patches on the skin **caused by years of sun exposure**. They are considered pre-cancerous, meaning **they can turn into a serious cancer if ignored**.
5. **Benign Keratosis-like Lesions (bkl):** Often called "age spots," these are **harmless crusty patches that appear on older adults**. Clinically, they are important because **they can easily be mistaken for melanoma by an untrained eye**.
6. **Vascular Lesions (vasc):** These are **harmless bumps caused by burst or enlarged blood vessels**, such as "cherry angiomas." They are usually bright red or purple.
7. **Dermatofibroma (df):** These are **harmless, firm, round bumps** that often form after a minor skin injury, like a bug bite.

While a trained dermatologist can usually tell these seven classes apart using visual patterns, it is a highly complex task. A harmless age spot (bkl) can look terrifyingly similar to a deadly melanoma (mel). To make matters worse, the dataset is heavily skewed toward harmless moles.

The Severity of Class Imbalance

In a **perfect experimental world**, a dataset would have exactly the same number of images for each disease. If we had **1,000 images of ordinary moles, we would also have 1,000 images of melanoma, and 1,000 images of Dermatofibroma**. This would **allow a neural network to learn the visual features of every disease equally well**.

However, medical datasets reflect clinical reality. **Doctors see thousands of ordinary moles for every rare disease they encounter. Because the HAM10000 dataset was collected from real clinics, it suffers from extreme class imbalance.**

To illustrate this, look at the distribution of images in our training data:

Table 3.1: Training Set Class Distribution

Disease Class	Medical Name	Image Count	Status
nv	Melanocytic Nevi	4,683	Extreme Majority
bkl	Benign Keratosis	772	Minority
mel	Melanoma	773	Minority
bcc	Basal Cell Carcinoma	361	Minority
akiec	Actinic Keratoses	222	Severe Minority
vasc	Vascular Lesions	99	Severe Minority
df	Dermatofibroma	71	**Extreme Minority**

As the table shows, **the network will look at 4,683 pictures of ordinary moles (nv) but only 71 pictures of Dermatofibroma (df). This creates a ratio of roughly 70 to 1.**

We deliberately chose to focus our study on the df class because it represents the worst-case scenario. It is an ablation study for extreme minority recovery. If we can successfully design an artificial intelligence pipeline that learns the true boundaries of a disease from just 71 images—without degrading its ability to recognize the thousands of ordinary moles—then we have proven that our methodology works.

The Hidden Trap: Data Leakage

Before we can train a model to solve this imbalance, we have to prepare the data. **In machine learning, you never train a model on all of your data.** If you do, **you have no way of knowing if the model actually learned general concepts, or if it simply memorized the answers.**

To solve this, **researchers split their data. A standard split might put 70% of the images into a "Training Set" (which the model learns from), 15% into a "Validation Set" (which is used to tune the model during training), and 15% into a "Test Set" (which is locked away and only used at the very end to evaluate the final model).**

Usually, researchers just shuffle their images and randomly divide them into these three buckets. However, in medical imaging, randomly splitting images is a dangerous mistake. It causes a critical failure known as data leakage.

To understand data leakage, imagine you are a medical student taking a final exam. The night before the test, your professor accidentally emails you the answer key. When you take the exam the next day, you score 100%. On paper, you look like a genius. But in reality, you did not learn medicine; you simply memorized the test.

Neural networks are incredibly lazy. They will take any shortcut they can find to minimize their mathematical error. In the HAM10000 dataset, there is a hidden structural trap. The dataset contains two different tracking numbers for every photograph: an `image_id` and a `lesion_id`.

The `image_id` tracks the specific photograph. The `lesion_id` tracks the actual, physical bump on the patient's skin.

When a patient visits a dermatologist, the doctor rarely takes just one photograph. They might take a picture from the left side, another picture from the right side, and a third picture with different lighting. Therefore, a single physical skin lesion (`lesion_id`) might have three different photographs (`image_id` A, B, and C).

If a researcher ignores the `lesion_id` and randomly splits the data using the `image_id`, a catastrophic leak occurs.

Imagine Patient John has a melanoma. The doctor took three photos of it. In a random split, Photo A and Photo B might end up in the Training Set, while Photo C ends up in the Test Set.

During training, the neural network studies Photo A and Photo B. It looks very closely at the exact shape of John's tumor, the specific color of his surrounding skin, and even the hair follicles nearby. It memorizes the image.

Later, during the final evaluation, the network is shown Photo C from the Test Set. The network is supposed to be looking at a completely unseen cancer to prove it understands what melanoma looks like in general. Instead, it recognizes John's skin. It recognizes the exact same hair follicles. It immediately outputs "Melanoma."

The researcher sees that the model predicted the Test Set image correctly and celebrates a high accuracy score. But the model is an illusion. It did not recognize the disease; it recognized the patient. When this model is deployed in

a real hospital and shown a completely new patient, it will fail.

The Solution: Lesion-Level Stratified Splitting

To ensure that our experimental results are scientifically honest, we had to build a strict data integrity framework to prevent leakage. We discarded the standard random image split and built a custom Lesion-Level Stratified Split.

We executed this in three steps:

Step 1: Isolate Unique Lesions Instead of looking at the 10,000 individual images, we grouped the dataset by `lesion_id`. We told the computer to look only at the physical skin bumps, ignoring how many photos were taken of each one.

Step 2: Stratified Lesion Splitting We then randomly divided the lesions into our three buckets (70% Train, 15% Validation, 15% Test).

We also used a statistical technique called "stratification" during this split. **Stratification ensures that the percentages of the diseases remain balanced across all three buckets.** For example, if we did a purely random split, we might accidentally put all 70 of the rare Dermatofibroma lesions into the Test Set, leaving zero for the Training Set. **Stratification guarantees that 70% of the `df` lesions go to the Train Set, 15% go to the Val Set, and 15% go to the Test Set.**

Step 3: Map Back to Images Once the physical lesions were securely sorted into their separate buckets, we mapped them back to their individual photographs. If Patient John's melanoma was assigned to the Training Set, all three of his photographs were sent to the Training Set.

This guarantees a strict firewall. **The Validation Set and Test Set now contain completely unseen patients and completely unseen lesions.** If our model successfully diagnoses an image in the Test Set, it is because it genuinely learned the biological patterns of the disease, not because it memorized a patient's skin.

After applying this strict splitting logic, our final dataset was safely divided.

Table 3.2: Final Dataset Split Counts

Split Name	Image Count	Purpose
Train Set	6,981	Used to teach the neural network.
Validation Set	1,532	Used to evaluate the model during training and adjust internal settings.

| ****Test Set**** | 1,502 | Locked away. Used only once to report final performance. |

With the images securely isolated, we applied **standard medical image preparation techniques. We resized every image to a standard 128x128 pixel resolution so the neural network would process them uniformly, and we normalized the pixel colors to ensure lighting differences across clinics did not confuse the model.**

Why Data Integrity Must Come First

In the machine learning community, there is an intense focus on writing complex code, designing deeper neural networks, and inventing new mathematical loss functions. However, none of that matters if the data pipeline is flawed.

A model evaluated on leaked data will produce artificially high metrics. If we had allowed data leakage in this project, we would never have seen the model fail on the minority class. The network would have simply memorized the 70 df images, scored highly on the test set, and falsely convinced us that extreme class imbalance was not a problem.

By building a strict, trustworthy data pipeline, we created a harsh but honest environment. We can now trust our metrics entirely. When a model fails in the next chapter, we will know it is a genuine mathematical failure caused by class imbalance, not a bug in our code. And when a model finally succeeds later in the project, we will know the success is real.

Chapter Summary

Medical datasets like HAM10000 suffer from severe class imbalance. While ordinary moles are represented by thousands of images, rare conditions like Dermatofibroma are represented by less than a hundred. To test how to solve extreme class imbalance, we selected the Dermatofibroma class as our primary focus. Before building any models, we identified a critical danger in medical data processing: data leakage. Because dermatologists take multiple photos of the same physical lesion, standard random splitting allows the model to memorize patients rather than learn diseases. By isolating unique lesions and performing a stratified lesion-level split, we built a strict firewall between our training, validation, and testing sets. With this trustworthy foundation established, we are now ready to train our first neural network and observe exactly how standard algorithms fail when faced with imbalanced data.

Chapter 4: Baseline Experiment – Understanding Failure

In the scientific method, before you attempt to cure a disease, you must first observe it in its natural state. If we immediately applied advanced generative artificial intelligence to our dataset, we would not have a reference point. If the final system achieved a high score, we would not know if the generative model actually helped, or if a simple network could have achieved the exact same result on the raw data.

To measure progress, we must first establish a baseline. **A baseline is a standard, unmodified machine learning model trained on the original dataset without any special tricks to fix the class imbalance. By training a baseline model, we can observe exactly how a neural network naturally behaves when it is forced to look at 4,683 pictures of ordinary moles and only 71 pictures of the rare Dermatofibroma.**

This chapter explains how we built our baseline model, how we evaluated it, and why its ultimate failure was both mathematically predictable and highly educational.

Why ResNet18 and Transfer Learning

When building an image classification system, we do not write the code from scratch. Instead, we use established architectures that have proven successful in the computer vision community. For this project, we selected ResNet18.

As we discussed in the previous chapter, **ResNet18 is a Convolutional Neural Network with 18 layers and residual skip connections.** We selected this specific architecture because it strikes the perfect balance between learning capacity and simplicity. It is deep enough to understand complex medical textures, but small enough that it will not immediately memorize our tiny minority classes.

However, even a small model like **ResNet18 contains over 11 million learnable parameters** (the internal mathematical weights the network adjusts during training). **If we tried to train 11 million parameters using only the 6,981 images in our training set,** the network would struggle to learn anything meaningful. It would be like asking a child to learn how to read an entire language by showing them only a single short book.

To solve this, we used a strategy called **transfer learning.**

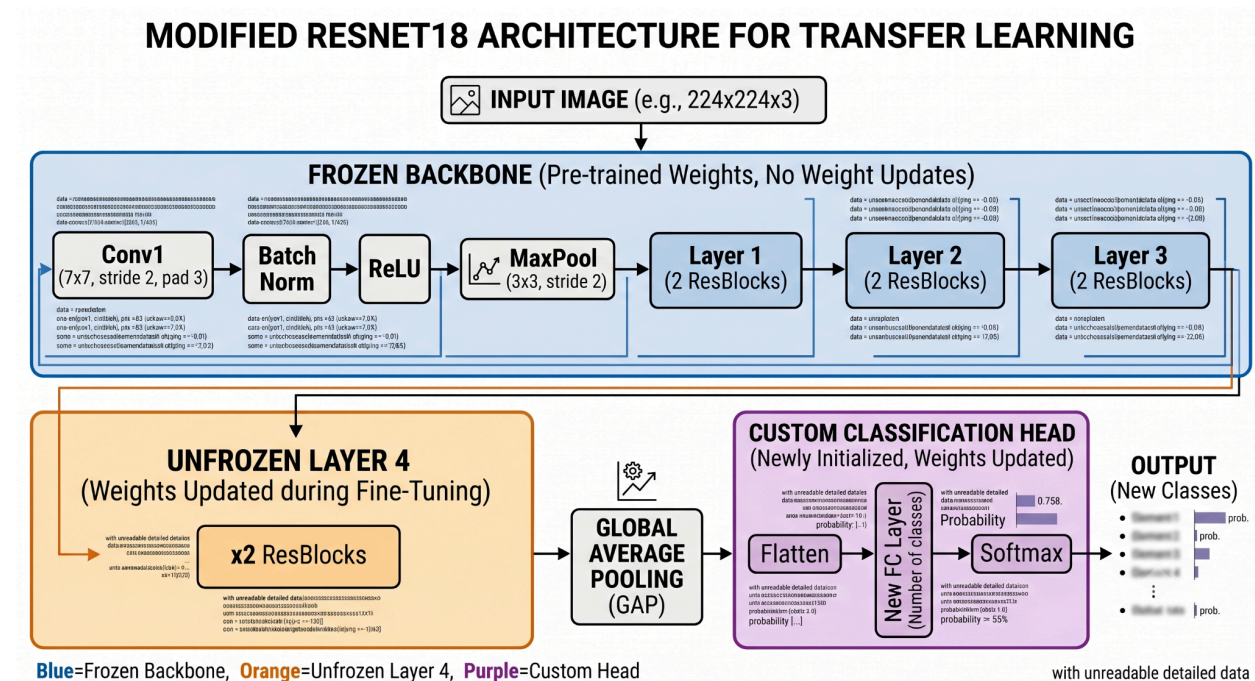
Instead of starting with a blank, untrained network, **we downloaded a version of ResNet18 that had already been trained on ImageNet,** a massive dataset containing over a million pictures of everyday objects like cats, dogs, cars, and trees. **When a network is trained on a million images, its early layers learn universal visual concepts. They learn how to detect straight edges, curved lines, shadows,**

and color contrasts.

By using this pre-trained network, we gave our model a massive head start. The network already knew how to "see." We just needed to teach it the specific vocabulary of skin cancer.

Model Architecture and Modifications

We could not simply drop our medical images into the pre-trained ResNet18 model and expect it to work. **The original network was designed to output 1,000 different classes of everyday objects. We had to modify it to output exactly 7 classes of skin lesions.**



[Placeholder: Figure showing the modified ResNet18 architecture, highlighting the frozen backbone, the unfrozen Layer 4, and the custom classification head.]

First, we locked the mathematical weights in the earliest layers of the network. This is called "freezing" the backbone. Because the early layers detect basic things like lines and shadows, we did not want the network to accidentally forget these universal concepts during training.

Second, we left the final convolutional section of the network, known as "Layer 4," completely unlocked and trainable. This is the section of the network responsible for piecing together simple lines into complex shapes. By keeping Layer 4 unfrozen, we allowed the network to learn the specific, highly complex spatial boundaries and pigment networks unique to dermatology.

Finally, **we chopped off the network's original classification head and replaced it with a custom head designed specifically for our project.** We passed the visual information through a new hidden layer of 128 neurons. More importantly, we added heavy "Dropout" layers. **Dropout is a technique where the network randomly turns off a percentage of its neurons during every training step.** Imagine a student studying for an exam. If the student always studies with the exact same group of friends, they might just rely on the smartest friend for the answers. If the teacher randomly removes a different friend from the study group every day, the student is forced to actually learn the material themselves. Dropout does the same thing: it prevents the network from relying on memorization and forces it to learn robust, generalizable medical features.

The Training Pipeline

With the architecture built, we set up the training pipeline. **The network was trained to look at batches of 32 images at a time. It made predictions, checked its answers against the true labels, and adjusted its internal weights to become more accurate.**

To guide the network, we used a mathematical formula called **Standard Cross-Entropy Loss**. A loss function is simply the way we measure how badly the network messed up. **Standard Cross-Entropy treats every mistake equally. If the network misclassifies an ordinary mole, the loss goes up. If the network misclassifies a rare Dermatofibroma, the loss goes up by the exact same amount.** The network's only goal is to make the total loss as small as possible.

We optimized the network using an algorithm called AdamW. This acts as the steering wheel during training, helping the network adjust its weights efficiently without making erratic jumps. **If the network stopped improving for several rounds, we automatically reduced the learning speed, allowing it to fine-tune its understanding gently.**

The Danger of Accuracy

Before we look at the results of this baseline experiment, we must discuss how we evaluate the model. In general computer science, if a program is correct 95 percent of the time, we consider it a massive success. **In medical artificial intelligence, focusing only on this overall accuracy is one of the most dangerous mistakes a researcher can make.**

To understand why, imagine an airport security guard trying to find smuggled items in luggage. Suppose 99 out of 100 bags contain perfectly legal clothing, and only 1 bag contains a dangerous weapon.

If the security guard simply waves every single person through the scanner without checking a single bag, they will be correct 99 times out of 100. Their overall accuracy is 99 percent. If we only look at accuracy, we will promote the guard and call the system perfect. But the system is completely broken. It failed entirely at the one task it was designed to do: finding the dangerous item.

This is exactly what happens when neural networks process imbalanced medical datasets. **Because standard cross-entropy loss treats all mistakes equally, the network realizes that the easiest way to minimize its error is to just vote for the majority class every single time. It acts exactly like the lazy security guard.**

To expose this lazy behavior, we must use stricter evaluation metrics. Instead of looking at overall accuracy, we break the performance down into three deeper concepts: Precision, Recall, and the F1-Score.

Recall answers a critical medical question: Out of all the patients who actually have a specific disease, how many did the model successfully find? If there are 20 patients with Dermatofibroma in the test set, and the model only flags 10 of them, the recall is 0.50 (or 50 percent). In healthcare, maximizing recall is highly desirable because missing a disease can cost a patient their life.

Precision answers a different question: Every time the model claims a patient has a disease, how often is it actually correct? **If the model flags 100 patients for Dermatofibroma, but only 10 of them actually have it, the precision is 0.10 (or 10 percent). Low precision means the system is generating a massive number of false alarms. While false alarms are generally better than missing a disease entirely, a system with terrible precision is useless in a real hospital because it will waste doctors' time and cause patients unnecessary panic.**

The F1-Score is simply the mathematical balance between Precision and Recall. It ensures that a model cannot cheat by just maximizing one metric while destroying the other.

Finally, to judge the entire system fairly, we use a metric called Macro F1. To calculate the Macro F1, we calculate the F1-Score for all seven diseases separately, and then we take the exact average of those seven scores. Macro F1 treats the rare 70-image disease as mathematically equal to the common 6,700-image disease. It is the ultimate test of fairness. If a model completely fails on a rare disease, the Macro F1 score will plummet, even if the overall accuracy remains high.

Baseline Results

After training the ResNet18 model on our strictly separated training set, we locked the network and tested it on the 1,502 unseen images in our Test Set.

The results were precisely as mathematical theory predicted.

Table 4.1: Experiment 1 (Baseline ResNet18) Test Set Results

- **Overall Accuracy: 0.7763**
- **DF Precision: 0.0000**
- **DF Recall: 0.0000**
- **DF F1-Score: 0.0000**
- **Macro F1-Score: 0.5307**

On the surface, the model achieved an accuracy of 0.7763. It correctly diagnosed over 77 percent of the images it saw. To an untrained observer, this might look like a highly successful first attempt.

However, when we look at the metrics **for the rare Dermatofibroma (DF) class, we see absolute failure. The DF Precision is zero. The DF Recall is zero. The DF F1-Score is zero. Out of the 20 actual Dermatofibroma images hidden inside the Test Set, the model found absolutely none of them. Furthermore, the Macro F1-score sits at a very low 0.5307, exposing the severe imbalance in the system's knowledge.**

Analyzing the Confusion Matrix

To understand exactly what the model was doing, we can look at its Confusion Matrix.

[Placeholder: Figure showing the 7x7 Baseline Confusion Matrix]

A confusion matrix is a grid that shows exactly where the model made its mistakes. The true diseases are listed on one side, and the model's predictions are listed on the other. If the model is perfect, all of the numbers will sit perfectly on the diagonal line running down the middle of the grid. Any numbers falling outside that diagonal line represent confusion.

When we examine the confusion matrix for the baseline model, the behavior is glaringly obvious. **The model became highly confident at identifying Nevus (ordinary moles).** Whenever it was unsure about an image, it simply guessed Nevus. It essentially acted as a statistical voting machine. Because it saw Nevus thousands of times during training and Dermatofibroma only 71 times, **it mathematically learned that guessing the rare disease was not worth the risk.**

The model did not fail to learn. It learned exactly what we programmed it to learn. We programmed it to use Standard Cross-Entropy loss, which minimizes average error. By ignoring the rare class and mastering the common class, the network successfully minimized its average error and achieved nearly 78 percent accuracy.

What We Learned

This baseline experiment proves a foundational concept in medical machine learning: you cannot trust standard overall accuracy. If we had only looked at the top-line accuracy score, we would have believed our model was performing well. By looking at the strict precision and recall metrics for the minority classes, we exposed the illusion.

We learned that a standard CNN, no matter how powerful its architecture, will always take the path of least resistance. It will succumb to the bias of the data.

To build a reliable medical diagnostic tool, we must intervene. **We have to change the rules of the training process so that the network is no longer allowed to ignore rare diseases safely. We need to find a way to make mistakes on the minority class mathematically painful.**

This naturally leads us to our next question: **What happens if we intervene in the loss function? If Standard Cross-Entropy allows the network to ignore Dermatofibroma, what happens if we assign a massive mathematical penalty every single time the model gets a Dermatofibroma wrong? Will punishing the network solve the problem, or will it create an entirely new type of failure? We will test this intervention in the next chapter.**

Chapter Summary

In this chapter, we established a baseline model to observe how a neural network naturally handles severe class imbalance. We used a pre-trained ResNet18 architecture to leverage transfer learning, locking the early layers to retain basic visual knowledge while leaving the deeper layers open to learn complex skin cancer features. We trained the model using Standard Cross-Entropy loss, which treats every mistake equally. While the resulting model achieved an overall accuracy of 77.63 percent, deeper evaluation metrics revealed a severe failure. The model completely ignored the rare Dermatofibroma class, resulting in zero precision and zero recall. This experiment proved that overall accuracy is an illusion in imbalanced medical datasets. The network took the path of least mathematical resistance, minimizing its average error by simply guessing

the majority class. This failure proves that standard training is insufficient and motivates our first mathematical intervention: weighted loss.

Chapter 5: Experiment 2 – Weighted Loss: Making Rare Mistakes Expensive

At the end of the previous chapter, we exposed a critical flaw in our baseline model. Despite achieving an overall accuracy of nearly 78 percent, the neural network completely ignored the rare Dermatofibroma class. **It scored a zero in both precision and recall.** This failure left us with a highly specific question: **If the model ignores a rare disease because making a mistake carries almost no mathematical penalty, what happens if we force those mistakes to become incredibly expensive?**

In this chapter, we test our first major intervention. We will explain how we changed the rules of the training process, how we calculated specific penalties for different diseases, and why this experiment serves as a crucial scientific control. Finally, we will look at the results to understand why forcing a model to pay attention improves fairness, but simultaneously reveals the limits of mathematics when visual data is missing.

Why Standard Cross-Entropy Failed

To fix a problem, we must first understand exactly how it works. During our baseline experiment, we trained the ResNet18 model using a formula called Standard Cross-Entropy Loss.

In machine learning, the "loss" is simply the measurement of error. The network's only goal in life is to make this error number as small as possible. Standard Cross-Entropy computes this error by looking at every image in a training batch, checking if the network guessed correctly, and handing out a standard penalty for every wrong guess.

Because our training dataset contains 4,683 images of ordinary moles (Nevus) and only 71 images of Dermatofibroma, the mathematics become heavily skewed. If the network looks at 100 random images, roughly 67 of them will be ordinary moles, and only 1 might be a Dermatofibroma.

If the network decides to completely ignore the Dermatofibroma and simply guess "ordinary mole" every time, it gets 67 questions right and 1 question wrong. The **Standard Cross-Entropy formula gives it 1 unit of penalty.** The network is happy. It has successfully minimized its error with almost no effort. It has no

mathematical incentive to learn the complex visual patterns of a disease that barely ever appears.

The Intuition Behind Weighted Loss

If we cannot change the number of images we have—at least not yet—how do we stop the network from taking this lazy shortcut? We do it by changing the grading rubric.

Imagine a medical student taking a final exam consisting of 100 questions. Ninety-nine of these questions are about basic anatomy. One question is about how to safely administer a highly dangerous drug.

If the professor grades the exam so that every question is worth exactly 1 point, a student could entirely skip the drug safety question and still score a 99 percent on the exam. They would pass with flying colors, despite missing a piece of critical, life-or-death knowledge. This is exactly how Standard Cross-Entropy works.

Now, imagine the professor changes the rules. The 99 anatomy questions are still worth 1 point each. But the single drug safety question is now worth 100 points. If the student skips the safety question, they immediately fail the exam, regardless of how well they know basic anatomy. The student is now forced to study the safety protocol just as intensely as the rest of the material.

This is the exact intuition behind Weighted Loss. Instead of handing out a standard penalty for every mistake, we assign a massive mathematical penalty when the network misclassifies a rare disease, and a very small penalty when it misclassifies a common disease.

Calculating the Class Weights

To implement Weighted Loss, we need to calculate exactly how much each disease should be penalized. We cannot just guess these numbers; we must calculate them based on the actual distribution of our data.

The goal of these weights is to mathematically balance the dataset. We want the total penalty of all 71 Dermatofibroma images to equal the total penalty of all 4,683 Nevus images. **To do this, we use a technique called "inverse frequency weighting."**

The formula simply takes the total number of images in the dataset and divides it by the number of images in a specific class, adjusted by the number of total classes. If a class has very few images, the math produces a very large weight.

If a class has many images, the math produces a very small weight.

Applying this formula to our HAM10000 training set resulted in the following class weights:

Table 5.1: Applied Class Weights for Experiment 2

Disease Class	Medical Name	Image Count	Calculated Weight
-----	-----	-----	-----
df	Dermatofibroma	71	**14.0463**
vasc	Vascular Lesions	99	10.0736
akiec	Actinic Keratoses	222	4.4923
bcc	Basal Cell Carcinoma	361	2.7626
bkl	Benign Keratosis	772	1.2918
mel	Melanoma	773	1.2901
nv	Melanocytic Nevi	4,683	**0.2130**

This table clearly illustrates the massive shift in incentives. If the neural network misclassifies an ordinary mole (nv), it receives a tiny penalty of 0.21. But if it misclassifies a Dermatofibroma (df), it is hit with a massive penalty of 14.04. A mistake on the rarest class is now mathematically equivalent to missing roughly 66 ordinary moles. The network can no longer afford to be lazy.

A Necessary Scientific Control

At this point in the project, we must pause to answer an important scientific question: Why are we doing this experiment? Why did we not immediately skip to generating new images with artificial intelligence?

In scientific research, you must change only one variable at a time. If we immediately applied a Generative Adversarial Network (GAN) to our dataset and the model performance improved, we would have a fatal gap in our knowledge. A critic could point out that we never tested whether a simple mathematical formula could have solved the problem instead. We would have no way to prove that the complex artificial intelligence was actually necessary.

Experiment 2 acts as our scientific control. Weighted Loss changes the optimization landscape. It changes the incentives, the penalties, and the way the network updates its internal parameters. **However, it is absolutely critical to understand that Weighted Loss does not create new information.**

The network is suddenly forced to pay deep attention to Dermatofibroma, but it still only has 71 photographs to look at. By running this experiment, we can isolate exactly how much of the class imbalance problem can be solved by

mathematics alone, before we alter the actual visual data.

The Results: Mathematics Forces Attention

We took the exact same ResNet18 architecture from Experiment 1, with the exact same frozen backbone and trainable Layer 4. We trained it on the exact same dataset splits. The only thing we changed was the loss function, plugging in the weights from Table 5.1.

After training, we evaluated the model on the locked Test Set. The results showed a dramatic behavioral shift.

Table 5.2: Comparing Baseline vs. Weighted Loss Results

Metric	Experiment 1 (Baseline)	Experiment 2 (Weighted Loss)	Change
Overall Accuracy	0.7763	0.7417	Dropped by ~0.03
DF Precision	0.0000	0.5000	Massive Increase
DF Recall	0.0000	0.5000	Massive Increase
DF F1-Score	0.0000	0.5000	Massive Increase
Macro F1-Score	0.5307	0.6019	Improved by ~0.07

The intervention worked. By making rare mistakes expensive, we successfully woke the network up.

In the baseline experiment, the network found zero Dermatofibromas. In this experiment, the DF Recall jumped from 0.0000 to 0.5000. Out of all the actual Dermatofibromas hidden in the Test Set, the model successfully identified exactly half of them.

Furthermore, the DF Precision also jumped to 0.5000. This means that whenever the model flagged an image as a Dermatofibroma, it was correct 50 percent of the time. While not perfect, this is an astronomical improvement over the baseline, where the model was too afraid to even attempt a prediction.

Most importantly, the Macro F1-Score—our ultimate measure of system-wide fairness and balance—rose from 0.5307 to 0.6019. The model became a significantly more balanced diagnostic tool across all seven diseases.

Why Did Overall Accuracy Decrease?

A careful observer will notice an apparent contradiction in the results table. If the model became so much better at finding rare diseases, why did the overall accuracy fall from 77.63 percent to 74.17 percent?

This drop in accuracy is not a failure; it is a mathematically expected tradeoff.

In Experiment 1, the network achieved a falsely high accuracy by relying entirely on the majority class. It guessed "ordinary mole" almost every time, and because the test set is dominated by ordinary moles, it was usually right.

In Experiment 2, the severe penalties forced the network to change its strategy. Because missing a rare disease is now so mathematically painful, the network becomes much more cautious. If it sees an image that is ambiguous—perhaps a mole with a slightly rough edge—it will take a risk and guess that it might be a rare disease.

By taking more risks, the model successfully catches more real cases of Dermatofibroma (increasing recall). However, because it is taking more risks, it also accidentally flags some harmless moles as rare diseases. Every time it falsely flags a harmless mole, it gets a question wrong, and the overall accuracy metric takes a hit.

This tradeoff highlights exactly why accuracy is a dangerous metric in healthcare. We willingly sacrificed 3 percent of our overall accuracy to ensure that the system no longer ignores a rare disease entirely. **In a clinical setting, a system that catches half of the rare cases but occasionally raises a false alarm is vastly superior to a system that misses every single rare case while boasting a high accuracy score.**

Analyzing the Confusion Matrix

To see this risk-taking behavior in action, we turn to the confusion matrix.

[Placeholder: Figure showing the 7x7 Weighted Loss Confusion Matrix]

When we compare this matrix to the baseline, the changes are immediate. In the baseline matrix, the column for Dermatofibroma was completely empty. The model placed zero bets on the rare class.

In the Weighted Loss confusion matrix, the model is actively placing bets. The diagonal line, which represents correct predictions, is much thicker across all the minority classes. The model is successfully identifying cases of Dermatofibroma, Vascular Lesions, and Actinic Keratoses that it completely missed before.

However, the matrix also reveals the cost of this success. We see more numbers

scattering outside the diagonal line. Specifically, we see harmless Nevus images being incorrectly sorted into the minority class columns. The model has become highly sensitive, leading to a rise in false positives. The network is trying its best to balance the massive penalties, but it is struggling.

Strengths, Limitations, and the Core Insight

Experiment 2 was a success in terms of proving a concept. **It proved that a neural network's bias toward majority classes can be mitigated by manipulating the loss function.** The strength of Weighted Loss is its simplicity. It requires no extra data collection, no complex image generation, and very little computational overhead. By simply changing a few lines of code to update the mathematical grading rubric, we dramatically improved the system's Macro F1 balance.

However, the confusion matrix also revealed the absolute limits of this approach.

While the model is now paying deep attention to Dermatofibroma, it still struggles to separate the rare disease perfectly from common moles. Why is this happening?

The answer lies in the fundamental difference between incentives and information.

Weighted loss changed the model's incentives. It forced the neural network to stare very, very closely at the 71 images of Dermatofibroma. But staring harder at a small amount of data does not magically create new information. The network still only knows about 71 specific shapes, 71 specific lighting conditions, and 71 specific pigment patterns.

When the model is presented with a new test image that looks slightly different from those 71 examples, it panics. It knows that a mistake will result in a massive 14.04 penalty, but it simply does not have enough visual experience to know exactly where the boundary between a healthy mole and a Dermatofibroma lies.

This brings us to the key insight of the project so far: **Mathematical optimization can force a neural network to care about a rare class, but it cannot teach the network what the rare class actually looks like in all its diverse forms. To improve the model's confidence and reduce its false alarms, we must move beyond manipulating the loss function. We must provide the network with more visual diversity.**

If we do not have enough real photographs, the logical next step is to artificially create new ones. Before we jump to complex artificial intelligence generators, we must first test the standard industry approach. What happens if we take our 71 images and artificially expand them by rotating them, flipping them upside down, and zooming in? Can classical geometric tricks provide the visual diversity the network is so desperately missing?

This question drives us directly into our next intervention, testing the limits of classical data augmentation.

Chapter Summary

In this chapter, we addressed the complete failure of the baseline model by implementing our first scientific control: Weighted Loss. Because Standard Cross-Entropy allows a network to minimize its error by ignoring rare classes, we calculated inverse-frequency class weights to drastically increase the penalty for misclassifying minority diseases. A mistake on a rare Dermatofibroma was mathematically weighted to be 66 times more painful than a mistake on a common mole. This intervention successfully forced the model to pay attention, raising Dermatofibroma recall from zero to 0.5000 and pushing the Macro F1-Score to 0.6019. The overall accuracy dropped to 74.17 percent, a natural tradeoff caused by the model taking more risks to find the rare disease. However, analyzing the confusion matrix revealed that while the network was trying to learn, it was fundamentally limited by a lack of visual information. Mathematical penalties change a network's incentives, but they do not provide new visual examples. To give the network the morphological diversity it needs to confidently draw decision boundaries, we must alter the data itself. This insight motivates our next experiment: attempting to expand the dataset using classical image augmentation.

Chapter 6: Experiment 2.5 – Classical Augmentation: More Images, Same Knowledge?

At the end of our previous chapter, we successfully forced the neural network to pay attention to the rare Dermatofibroma class. By changing the mathematical grading rubric to heavily penalize mistakes on rare diseases, we raised the network's recall from zero to fifty percent. However, we also exposed a severe limitation. Weighted loss changed the network's incentives, but it did not create new information. The network was trying its hardest to identify the rare disease, but it was still trying to learn the boundaries of a complex medical condition using only 71 photographs.

If the model is starving for visual information, the most obvious question is: What happens if we artificially create more training images?

Before we turn to complex artificial intelligence generators, we must first test the standard, industry-accepted method for expanding datasets. In this chapter, we explore a technique known as **Classical Data Augmentation**. We will explain how it works, why it is a standard tool in computer vision, and why it ultimately fails to solve the problem of extreme medical class imbalance. We will see firsthand how forcing a network to look at thousands of twisted, rotated copies of the same 71 images creates a dangerous tradeoff between catching a disease and causing false alarms.

What is Classical Data Augmentation?

In human vision, if you look at a coffee mug, and then you turn your head upside down and look at the exact same coffee mug, your brain instantly knows it is still a coffee mug. Your brain naturally understands that the orientation of an object does not change what the object is.

Convolutional Neural Networks do not have this natural understanding. A neural network sees an image simply as a giant grid of numbers representing pixel colors. If you take a picture of a skin lesion that is sitting in the top-left corner of the grid, and you physically move the lesion to the bottom-right corner of the grid, the underlying numbers change completely. To an untrained neural network, this looks like a completely different photograph.

Classical data augmentation is a technique used to teach the network that these variations do not matter. Instead of just showing the network the original image once, we write a computer script that randomly alters the image slightly every single time the network sees it.

This process is widely used in computer vision because it artificially expands the size of the training dataset. If you have 1,000 images, and you randomly rotate them, flip them, and zoom in on them during training, the network will effectively see millions of unique pixel grids. This prevents the network from memorizing the exact pixels of the original 1,000 images, a process that improves the model's overall robustness.

The Augmentation Toolbox

For this experiment, we built a heavy augmentation pipeline. **Every time an image from the training set was passed to the neural network, it was randomly subjected to the following transformations:**

Horizontal and Vertical Flips: The computer acts as a mirror, flipping the image left-to-right or top-to-bottom. In dermatology, this is highly effective. A mole is still a mole whether the doctor took the picture from the patient's left side or right side. Skin has no strict "up" or "down."

Random Rotation: The image is randomly spun by up to 20 degrees in either direction. This simulates the natural reality of a busy clinic, where a doctor might hold the dermatoscope camera at a slightly different angle for each patient.

Random Translation: The image is shifted slightly up, down, left, or right. This teaches the network that a tumor does not have to be perfectly centered in the middle of the photograph to be dangerous.

Color Jitter: The computer randomly alters the brightness, contrast, and color saturation of the photograph. Clinics use different cameras, and rooms have different lighting—some use bright fluorescent lights, while others have natural sunlight. **Color jitter forces the network to focus on the structure of the disease rather than just the exact shade of the lighting.**

Affine Transforms (Scaling and Shearing): The image is slightly zoomed in or zoomed out, and stretched slightly along its axes. This simulates a doctor pressing the camera slightly harder against the skin, or photographing the skin from a slightly further distance.

[Placeholder: Figure showing a single Dermatofibroma image passing through the augmentation pipeline, resulting in six visually distinct, altered versions of the original lesion.]

Volume vs. Diversity: The Illusion of New Data

By combining our heavy class penalties from Experiment 2 with this new heavy augmentation pipeline, we created Experiment 2.5. We call it Experiment 2.5, rather than Experiment 3, because it acts as the final phase of our scientific control testing.

With this pipeline, the 71 images of Dermatofibroma are altered so heavily during training that the network will never see the exact same pixel grid twice. It will see tens of thousands of variations. It seems like we have solved the data shortage.

However, there is a fundamental difference between creating volume and creating diversity.

Imagine you want to teach a neural network to recognize dogs, but you only have one photograph of a Golden Retriever. You can rotate that photograph, flip it upside down, alter the lighting, and zoom in on its nose. You can create one million mathematically unique images from that single photograph. You have created massive volume.

But you have not created diversity. The network has still only ever seen a Golden Retriever. If you show the network a picture of a Bulldog, it will have no idea what it is looking at.

This is the exact problem with classical augmentation in extreme medical class imbalance. **Classical augmentation makes the network robust against camera angles, but it does NOT create genuinely new skin lesions. A rotated Dermatofibroma is still the exact same physical bump on the exact same patient's skin. It has the same internal core, the same specific borders, and the same biological features. The network is seeing more images, but it is not gaining any new medical knowledge about how the disease might look on a different patient.**

The Results of Experiment 2.5

We trained our ResNet18 model using both the weighted loss penalties and the heavy classical augmentation pipeline. Once the training was complete, we locked the network and tested it on our completely unseen Test Set.

The results exposed the exact mathematical limits of this approach.

Table 6.1: Comparing Baseline, Weighted Loss, and Classical Augmentation

Metric	Experiment 1 (Baseline)	Experiment 2 (Weighted Loss)	Exp 2.5 (Weighted Loss + Classical Aug)
Overall Accuracy	0.7763	0.7417	0.6844
DF Precision	0.0000	0.5000	0.2692
DF Recall	0.0000	0.5000	0.7000
DF F1-Score	0.0000	0.5000	0.3889
Macro F1-Score	0.5307	0.6019	0.5705

When we look at the Dermatofibroma (DF) Recall, the model appears to be an incredible success. The recall skyrocketed to 0.7000. Out of all the actual Dermatofibroma images hidden in the test set, the model successfully caught 70 percent of them. This is the highest recall we have seen so far in the project. The heavy augmentation pipeline clearly forced the network to memorize every

possible camera angle and lighting variation of those 71 original images.

But when we look at the rest of the table, the illusion shatters.

The DF Precision collapsed to 0.2692. The Overall Accuracy dropped to 0.6844. And most importantly, the Macro F1-Score—the ultimate measure of system-wide balance—dropped from 0.6019 back down to 0.5705.

To understand why a model can have such an incredibly high recall but simultaneously be considered a failure in balance, we must understand the Precision-Recall Tradeoff.

Understanding the **Precision-Recall Tradeoff**

In machine learning, Precision and Recall are often locked in a tug-of-war. If you push one metric too high without adding genuinely new information, the other metric will almost always collapse.

To understand this in simple language, **imagine a doctor who is absolutely terrified of missing a cancer diagnosis because the hospital will fire them if they make a mistake. (This represents our Weighted Loss, which heavily penalizes missing a Dermatofibroma).**

To ensure they never miss a cancer, the doctor adopts a new, extreme strategy: they begin diagnosing almost every single patient who walks through the door with cancer. If a patient has a slightly weird freckle, the doctor calls it cancer. If a patient has a harmless age spot, the doctor calls it cancer.

At the end of the year, we review the doctor's records. Out of the 100 people who actually had cancer, the doctor successfully caught 70 of them. Their Recall is 70 percent. The doctor celebrates.

But then we look at the other side of the records. To catch those 70 people, the doctor handed out cancer diagnoses to thousands of perfectly healthy people. Out of all the people the doctor claimed had cancer, only about 26 percent actually did. Their Precision is 26 percent.

This doctor is useless to the hospital. They have caused massive panic, wasted resources on unnecessary biopsies, and failed to correctly identify healthy patients.

This is exactly what our neural network did in Experiment 2.5. We call this the "Band-Aid Effect."

Because the network was terrified of the heavy loss penalty, and because we flooded it with thousands of rotated and twisted variations of the exact same 71 images, it became hypersensitive. The network memorized the textures of those 71 images so intensely that anytime it saw a test image containing a vaguely similar texture—even if it was a completely harmless mole—the network panicked and guessed "Dermatofibroma."

Analyzing the Collapsed Metrics

This hypersensitive paranoia explains every single dropped metric in our results table.

Why did the DF Precision collapse to 0.2692? Because the model generated a massive wave of false alarms. It flagged dozens of images as Dermatofibroma, but the vast majority of those guesses were incorrect.

Why did the Overall Accuracy drop to 68.44 percent? Because the dataset is mostly made up of ordinary moles (Nevus). When the network started panicking and falsely labeling ordinary moles as rare diseases, it got those guesses wrong. Because ordinary moles make up the largest portion of the test set, destroying performance on the Nevus class destroys the overall accuracy of the entire system.

Why did the Macro F1-Score drop to 0.5705? The F1-Score is the mathematical average of precision and recall. It is designed to punish models that exhibit extreme behavior. Because the precision collapsed so violently, the DF F1-Score dropped to 0.3889. When this terrible score was averaged with the rest of the classes, the system-wide Macro F1-Score dragged downward. The system lost its equilibrium.

Visualizing the Paranoia: The Confusion Matrix

If we look at the confusion matrix for this experiment, the network's paranoia is mapped out clearly in numbers.

[Placeholder: Figure showing the 7x7 Confusion Matrix for Experiment 2.5]

In the matrix, the column for Dermatofibroma is no longer empty, but it is overflowing with the wrong images. We can see a massive scatter of numbers falling outside the correct diagonal line. Specifically, large numbers of nv (ordinary moles) and bkl (benign keratosis) are being shoved into the df column.

The network does not know where the actual visual boundary between a harmless mole and a Dermatofibroma lies. Because classical augmentation did not show the

network any new patients with Dermatofibroma, the network could not learn the subtle differences in borders, colors, and shapes that separate the classes in the real world. When forced to make a choice, the network simply guessed the class that carried the highest mathematical penalty.

The Core Insight and the Path Forward

Experiment 2.5 was not a waste of time. In fact, it is one of the most educational experiments in this project. It proves a fundamental limitation of modern computer vision.

More images do not necessarily mean more information.

We threw the standard industry toolkit at this problem. We used a powerful ResNet18 architecture. We mathematically balanced the loss function. We artificially expanded the dataset using a heavy classical augmentation pipeline. And yet, the system failed to achieve balance. **Classical augmentation substantially improved minority-class recall, but it introduced a severe precision-recall tradeoff and reduced overall accuracy.**

The network now cares deeply about Dermatofibroma. It has seen thousands of augmented images of Dermatofibroma. But all of those images are still derived from the exact same 71 lesions. We have squeezed every single drop of mathematical value out of those 71 photographs. There is simply no more biological information to extract from them.

If we want the neural network to stop panicking and start making confident, highly precise boundaries between diseases, we have to show it something it has never seen before. We have to provide new morphological diversity. We need new borders, new internal structures, and new pigment networks.

Since we cannot go to a hospital and collect hundreds of new real photographs of this rare disease, we must turn to artificial intelligence to build them for us.

What if we could teach a separate AI model to understand the fundamental rules of what makes a Dermatofibroma a Dermatofibroma? What if we could use that model to hallucinate entirely new, synthetic lesions instead of merely rotating the existing ones?

This profound question shifts our project from traditional machine learning into the realm of **Generative Artificial Intelligence**. In the next chapter, we will leave our ResNet classifier behind for a moment, and attempt to build a machine that dreams of skin cancer.

Chapter Summary

In this chapter, we attempted to solve the network's lack of visual information by implementing classical data augmentation. We combined our weighted loss penalties with an aggressive pipeline that randomly rotated, flipped, zoomed, and color-shifted the training images. The goal was to artificially expand the dataset. While this technique makes models robust against camera angles, it does not create new biological information; a rotated image of a tumor is still the exact same tumor. The results perfectly demonstrated this limitation. The minority recall spiked to 0.7000, but the precision crashed to 0.2692, dragging the overall accuracy down to 68.44 percent and the Macro F1-Score down to 0.5705. Because the network was heavily penalized for missing the rare class but lacked diverse examples of how the disease actually looks on different patients, it became hypersensitive. It suffered from the "Band-Aid Effect," aggressively guessing the rare class and generating massive false alarms. This experiment proves that classical augmentation generates volume, but not the morphological diversity required to draw precise decision boundaries. To provide genuinely new visual structures without collecting new patient data, we must now transition to Generative Artificial Intelligence.

Chapter 7: Building a DCGAN for Extreme Data Scarcity

At the end of our previous chapter, we reached the mathematical limits of our dataset. We discovered that while classical data augmentation—rotating, flipping, and zooming—can create thousands of new images, it does not create genuinely new information. A rotated photograph of a tumor is still the exact same tumor. Because our neural network was heavily penalized for missing the rare Dermatofibroma class, but lacked diverse examples of how the condition actually looks on different patients, it became hypersensitive. It began hallucinating the rare class everywhere, destroying the system's precision and overall accuracy.

This left us with a profound question: What if we could create entirely new synthetic image samples instead of repeatedly transforming the same 71 photographs?

In a perfect world, if a machine learning model needs more data to learn a rare class, the researchers would simply go to a hospital and collect more photographs. In the reality of medical artificial intelligence, this is rarely possible. Collecting medical data is incredibly difficult. It requires strict privacy approvals, expensive equipment, and hundreds of hours of labor from

expert dermatologists who must manually review and label every single photograph. Furthermore, because Dermatofibroma is relatively rare compared to ordinary moles, a clinic might have to screen thousands of patients just to find a handful of new examples.

Because collecting real medical data was impossible for this project, we had to turn to artificial intelligence to build the data for us. In this chapter, we leave our classification model behind for a moment and enter the world of Generative Artificial Intelligence. We will explain how generative models work, why training them on extremely small datasets usually results in catastrophic failure, and the specific engineering interventions we used to force our system to successfully hallucinate synthetic visual diversity.

Section 7.1 – Understanding GANs

To create new images, we rely on a framework known as a Generative Adversarial Network, or GAN. To understand how a GAN works without relying on heavy mathematics, it is helpful to use a simple real-world analogy.

Imagine a master art forger and a police detective. The art forger has never seen a real painting by a famous artist, but they are trying to create a fake painting that looks exactly like one. The police detective is handed two paintings: one is a real masterpiece, and the other is the fake created by the forger. The detective's job is to guess which one is real and which one is fake.

In the beginning, the forger is terrible. They just throw random splashes of paint on a canvas. The detective easily spots the fake and arrests the forger. However, the forger learns from their mistakes. The next time, they use better colors. The detective still spots the fake, but it takes a little longer. As this game of cat-and-mouse continues for thousands of rounds, both sides improve. The detective becomes a world-class expert at identifying paint textures and brush strokes. To survive, the forger is forced to become so incredibly skilled that they begin painting images that are visually very difficult to distinguish from the original training set.

This is exactly how a GAN operates. It consists of two separate neural networks locked in an adversarial game.

The first network is the Generator. This is the art forger. It starts with a canvas of pure, random static noise, and it performs mathematical operations to mold that noise into an image.

The second network is the Discriminator. This is the police detective. It is fed images from the real dataset and images created by the Generator. Its only job

is to output a probability score: is this image real, or is it fake?

As these two networks train together, they push each other to improve. The Generator learns what visual textures, shapes, and borders make an image look visually plausible. The Discriminator learns what tiny artifacts make an image look artificial. Through this competitive training process, the Generator eventually learns to output synthetic visual diversity that captures the characteristics observed in the original training data.

Section 7.2 – Why We Selected DCGAN

Today, the field of artificial intelligence possesses generative models that are vastly more powerful than the original GAN. Researchers use StyleGAN to generate ultra-high-definition human faces, and they use Diffusion Models to generate highly complex artwork from simple text prompts. Given these powerful tools, why did we select an older architecture known as the Deep Convolutional GAN (DCGAN)?

The answer lies in the fundamental goal of our project. We are not trying to generate 4K-resolution images to fool a human dermatologist. Our goal is strictly functional: we need to generate enough morphological variation—varied borders, shapes, and color transitions—to help a downstream classifier draw better mathematical boundaries between classes.

More importantly, advanced models like StyleGAN and Diffusion Models are incredibly data-hungry. They require tens of thousands of high-quality images to understand how to map out complex visual rules. If we attempt to train a massive Diffusion Model on a tiny dataset of just 71 images, the massive network will simply memorize those 71 images and spit out exact copies, completely failing to provide the new variation we need.

DCGAN was selected because it is mathematically lightweight and architecturally stable. By utilizing standard convolutional layers rather than massive attention mechanisms, DCGAN requires significantly less computing power, making it highly practical for a constrained research environment. It is simple enough to be tightly controlled, which is an absolute necessity when dealing with the extreme constraints of our dataset.

Section 7.3 – The Reality of Training a GAN on Only 71 Images

In the computer vision industry, training a GAN on thousands of images is considered standard practice. Training a GAN on 71 images is considered an extreme scenario. When a researcher attempts to train a generative model on a "micro-dataset," the adversarial game between the Generator and the Discriminator usually breaks down immediately.

This breakdown happens in several ways, which we must understand before we look at our own training results.

The first major failure is Memorization, which is closely tied to Overfitting. Because the Discriminator only has 71 real images to look at, it does not learn general visual patterns. Instead, it just memorizes the exact pixel placements of those 71 specific photographs. Once it memorizes them, it rejects absolutely everything else as fake.

This leads directly to the second failure: Discriminator Domination. If the Discriminator perfectly memorizes the real images, the game becomes impossible for the Generator. No matter what the Generator draws, the Discriminator immediately flags it as fake with 100 percent confidence. When the Discriminator is perfectly confident, the mathematical feedback signal passing back to the Generator drops to zero. The Generator learns nothing. It is like a student taking a test, failing, and receiving a grade of zero with no feedback on what they did wrong. The student cannot improve.

The third failure is Mode Collapse. Sometimes, the Generator accidentally creates one single image that happens to fool the Discriminator. Because the Generator wants to win the game, it realizes that this one specific image is a guaranteed victory. It completely stops trying to learn how to draw diverse shapes or colors, and simply outputs that exact same image over and over again. The network has "collapsed" into a single mode of operation, completely destroying our goal of generating morphological variation.

Section 7.4 – Initial GAN Failure

Armed with an understanding of these expected failures, we began our first attempt at training the DCGAN. We isolated the 71 images of the Dermatofibroma class, resized them to 128 by 128 pixels, and fed them into the adversarial training loop.

We did not expect this first attempt to succeed, and the results proved us right. The training process collapsed almost instantly.

During the very first epochs of training, the Discriminator became vastly too strong. It perfectly memorized the 71 real images. Because the Discriminator became a perfect detective, the Generator's error rate exploded. The mathematical loss for the Generator shot up, while the Discriminator's loss dropped nearly to zero.

Because the Discriminator was rejecting everything with absolute certainty, the

Generator received no useful feedback. As a result, the Generator simply gave up. It produced screens of chaotic visual static and muddy, disorganized colors.

[Placeholder: Early GAN outputs showing severe static, disorganized colors, and signs of mode collapse]

This failure was not a mistake in the code; it was a fundamental reality of data scarcity. The detective was simply too smart, and the forger was trapped. If we wanted the Generator to actually learn how to hallucinate visually plausible lesion patterns, we had to intervene and forcefully handicap the Discriminator.

Section 7.5 – Stabilizing the GAN

To stabilize the adversarial game and prevent the Discriminator from dominating, we implemented four specific engineering interventions. These interventions altered the architecture and the rules of the training loop to force the two networks into a state of balance.

First, we introduced Discriminator Dropout. We refer to this technique as giving the Discriminator "amnesia." We inserted Dropout layers throughout the Discriminator's architecture, which randomly turned off 30 percent of its neurons during every single training step. By constantly disabling random parts of the Discriminator's brain, we made it physically impossible for the network to perfectly memorize the exact pixels of the 71 real images. The Discriminator was forced to look at general patterns, shapes, and color gradients rather than relying on photographic memory.

Second, we implemented One-Sided Label Smoothing. Normally, when we show the Discriminator a real image, we tell the computer that the image has a "Realness" score of 1.0. This allows the Discriminator to become absolutely certain of itself. To reduce this harmful confidence, we lied to the Discriminator slightly. Whenever we showed it a real image, we told it the image had a "Realness" score of 0.9. By removing the possibility of 100 percent certainty, we softened the mathematical gradients. This ensured that the feedback signal passed back to the Generator was gentle enough to learn from, rather than a harsh punishment.

Third, we utilized the Two Time-Scale Update Rule (TTUR). In a standard GAN, both the Generator and the Discriminator learn at the exact same speed. Because mapping real images is mathematically easier than creating fake images from scratch, the Discriminator naturally learns much faster. To fix this, we assigned completely different learning rates to the two networks. We restricted the Discriminator to a very slow learning rate of 0.00005, while we allowed the Generator to learn much faster at a rate of 0.00030.

Fourth, and finally, we implemented an Asymmetric Update Ratio. We changed the fundamental loop of the game. Instead of letting both networks take one turn at a time, we forced the Discriminator to wait. For every single time the Discriminator updated its weights, we allowed the Generator to update its weights three times. In our analogy, we allowed the art forger to practice painting three separate times before the detective was allowed to inspect their work.

By combining Dropout, Label Smoothing, TTUR, and the 3:1 Update Ratio, we effectively crippled the Discriminator just enough to give the Generator a fighting chance.

Section 7.6 – The Final Training Process

With our interventions in place, we restarted the training process. The dataset consisted exclusively of the 71 real Dermatofibroma images, normalized and scaled to a resolution of 128 by 128 pixels. We trained the networks for 1,000 full epochs.

[Placeholder: GAN Architecture Figure showing the Generator scaling up noise, and the Discriminator using convolutional layers with Dropout to output a probability score.]

This time, the training dynamics changed completely. The Generator and the Discriminator entered a healthy mathematical struggle. The Discriminator's confidence hovered around 50 percent, meaning it was genuinely confused and actively guessing whether the images were real or fake. This state of confusion provided rich, highly educational mathematical feedback to the Generator.

[Placeholder: Generator Loss Curve showing stable, oscillating error rather than explosive failure]

[Placeholder: Discriminator Loss Curve showing the network maintaining a healthy guessing state]

As the epochs progressed, we saved checkpoints of the Generator's internal weights and visually inspected the images it was producing. Around Epoch 200, the pure static began to consolidate into distinct circular shapes. By Epoch 500, the shapes developed fading borders that resembled the characteristics observed in the training data. By Epoch 1000, the Generator was consistently hallucinating visually plausible lesion patterns and background tones. The model had successfully learned to generate synthetic visual diversity based on the limited data.

Section 7.7 – The Target of 400 Synthetic Images

With the Generator successfully stabilized, we needed to decide how many synthetic images to create. We chose to generate exactly 400 new images.

This number was selected intentionally. If we generated 5,000 images, we might overwhelm the neural network and risk severe overfitting on artificial patterns. If we generated only 50 images, we would not solve the volume problem. By adding 400 synthetic images to our original 71 real images, the final training count for Dermatofibroma became 471.

This shifted the condition from an "extreme minority" class to a standard "minority" class, bringing its volume closer to other classes in the dataset like Basal Cell Carcinoma (361 images). This reduced the catastrophic 70:1 imbalance ratio down to a much more manageable 10:1 ratio against the majority class, providing enough volume for the classifier to learn without distorting the overall dataset geometry.

Section 7.8 – Evaluating the Synthetic Images

At this stage, we must be exceptionally clear and scientifically honest about how we evaluate the success of this GAN.

In mainstream computer vision research, the quality of generated images is usually evaluated using heavy mathematical formulas, the most common being the Fréchet Inception Distance (FID) or the Inception Score (IS).

In this project, no FID score or Inception Score was calculated.

This is not an oversight; it is a mathematical necessity. Formulas like FID rely on calculating massive statistical probabilities across thousands of samples. Attempting to calculate a meaningful FID score against a micro-dataset of only 71 images is mathematically unstable and scientifically meaningless. The resulting numbers would provide no actual proof of image quality.

Furthermore, we make no claims regarding the biological realism or medically validated structures of these synthetic images. We did not present these images to a board of human dermatologists to see if they could fool medical experts.

Instead, the quality of our GAN is evaluated strictly on two functional criteria.

First, we relied on visual inspection to verify that the Generator did not fall

into Mode Collapse. By inspecting batches of the generated images, we confirmed that the model was producing shapes with different background skin tones and different border variations. The network was successfully hallucinating morphological variation.

Second, and most importantly, the true evaluation of these synthetic images relies entirely on their downstream utility. We do not care if the images are beautiful art. We only care if they provide enough synthetic visual diversity to help a classification model draw better mathematical boundaries between classes. The true evaluation of this GAN will happen in the next chapter, when we feed these images into our ResNet18 classifier.

Section 7.9 – Key Lessons Learned

Training this generative model was a profound lesson in the realities of applied artificial intelligence. We learned that the standard tools of modern computer vision are highly brittle when deprived of massive data. Generative Adversarial Networks are incredibly powerful, but training them on a micro-dataset requires careful, deliberate engineering.

We learned that when a dataset is tiny, the Discriminator will always attempt to memorize the data rather than understand general patterns. To extract morphological variation from a handful of images, a researcher must intervene to slow the Discriminator down, blind its memory, and force a state of confusion. By applying Dropout, asymmetric learning rates, and unbalanced training loops, we successfully stabilized a chaotic mathematical system.

Through this struggle, we achieved our goal. The Generator network learned to approximate the characteristics observed in the training data. We used the finalized Generator to hallucinate 400 new, synthetic images of the class. We successfully created volume without resorting to classical augmentation.

The GAN has now done its job. We possess 400 synthetic images that feature morphological variation and synthetic visual diversity.

The most critical question of the entire project now remains: Can these hallucinated images actually improve a skin cancer classifier? Will this synthetic diversity finally stop our ResNet18 model from panicking and guessing randomly, or will it confuse the system even further?

To answer this, we must combine our real and synthetic data, and train our final Hybrid diagnostic model.

Chapter 8: Experiment 3 – The Hybrid Generative Pipeline

At the end of our previous chapter, we successfully trained a Deep Convolutional Generative Adversarial Network (DCGAN) to overcome the problem of extreme data scarcity. We bypassed the Discriminator's tendency to memorize the tiny dataset by deliberately slowing it down and giving it a kind of mathematical amnesia. In doing so, we forced the Generator to approximate the spatial patterns of the rare Dermatofibroma class. We used this stabilized Generator to create 400 new, synthetic images featuring morphological variation and synthetic visual diversity.

We now have 400 synthetic Dermatofibroma images. The critical question remains: Can these images actually improve a classifier?

In this chapter, we put our artificial data to the ultimate test. We will explain why generating images is completely useless unless those images improve a downstream task. We will construct a Hybrid Dataset, mixing our real photographs with our synthetic representations. Finally, we will test our ResNet18 classifier on this new dataset. By comparing the results against all our previous experiments, we will uncover a profound lesson about machine learning: success is not about maximizing a single metric like recall, but about restoring the delicate balance of the entire system.

The True Test of Synthetic Data

In the field of artificial intelligence, it is very easy to be distracted by beautiful pictures. Researchers often train generative models, print out a grid of the generated images, and declare the experiment a massive success simply because the images look visually plausible to the human eye.

However, looking plausible is not enough. If we generate 400 images of skin lesions that look realistic to a human, but those images contain hidden mathematical artifacts that confuse a neural network, then the generative model has failed. In applied machine learning, synthetic data has only one purpose: it must provide useful information to a downstream task.

In our project, that downstream task is our ResNet18 classification model. We do not care if our DCGAN won an art contest. We only care if the synthetic visual diversity it created can help the ResNet18 model draw improved decision boundaries between a harmless mole and a rare Dermatofibroma. If the classifier's observed downstream performance improves, the generative pipeline was a success. If the classifier's performance degrades, the pipeline was a

failure.

Constructing the Hybrid Dataset

To test the utility of our generated images, we had to carefully construct a new training environment. We created what we call a Hybrid Dataset.

We started with the 71 real images of Dermatofibroma from our original training set. Then, we injected the 400 synthetic images created by the DCGAN. This brought our final training count for the Dermatofibroma class to 471 images.

It is important to understand why we mixed the real and synthetic images together, rather than just training the model exclusively on the 400 synthetic images. If a neural network is trained only on synthetic images, it runs the risk of learning the specific mathematical quirks and imperfections of the Generator. By anchoring the synthetic images alongside the original 71 real photographs, we force the classification model to stay grounded in the real training data. The real images act as the foundation, while the synthetic images provide the morphological variation—the varied shapes, borders, and color distributions—that the network needs to understand how the class might look on a patient it has never seen before.

With 471 images, Dermatofibroma was no longer an extreme minority class suffering from a catastrophic 70:1 imbalance ratio. It was now roughly similar in size to other minority classes in the dataset, such as Basal Cell Carcinoma (which had 361 images). This provided enough volume for the network to learn, without completely distorting the natural geometry of the overall dataset.

Ensuring a Fair Scientific Experiment

In science, an experiment is only valid if you change exactly one variable at a time. If we changed our data and simultaneously upgraded our classification model to a massive, 152-layer network, we would have no idea which change actually caused the results.

To ensure strict fairness, we retained the exact same architecture we used in Experiment 1, Experiment 2, and Experiment 2.5. We used the same ResNet18 model. We kept the early backbone layers frozen, and we left Layer 4 and our custom classification head open to learn. We also evaluated the model on the exact same, strictly locked Test Set. The Test Set contained only real, unseen patients. No synthetic images were ever allowed near the Test Set.

The only thing that changed in Experiment 3 was the training data provided for the Dermatofibroma class, and a slight adjustment to the class penalties to

account for the new image count.

By keeping everything else perfectly still, any changes in the final metrics can be attributed entirely to the synthetic visual variation.

Final Results of the Hybrid Pipeline

We trained the ResNet18 model on the new Hybrid Dataset. We watched the loss curves stabilize, locked the network's weights, and ran it blindly against the Test Set.

When the metrics were calculated, they revealed a massive shift in the network's behavior. To understand the impact of the generative intervention, we must look at the results of Experiment 3 side-by-step alongside all of our previous attempts.

[Placeholder: Table 8.1 comparing the final metrics of all four experiments.]

Table 8.1: Master Results Comparison

Experiment 1: Baseline ResNet18 Accuracy: 0.7763 DF Precision: 0.0000 DF Recall: 0.0000 DF F1-Score: 0.0000 Macro F1-Score: 0.5307

Experiment 2: Weighted Loss Accuracy: 0.7417 DF Precision: 0.5000 DF Recall: 0.5000 DF F1-Score: 0.5000 Macro F1-Score: 0.6019

Experiment 2.5: Weighted Loss + Classical Augmentation Accuracy: 0.6844 DF Precision: 0.2692 DF Recall: 0.7000 DF F1-Score: 0.3889 Macro F1-Score: 0.5705

Experiment 3: DCGAN Hybrid Pipeline Accuracy: 0.7723 DF Precision: 0.4211 DF Recall: 0.4000 DF F1-Score: 0.4103 Macro F1-Score: 0.6166

When we first look at this table, the numbers might seem counterintuitive. If we focus entirely on minority recall, Experiment 2.5 appears to be the winner. In Experiment 2.5, the model successfully caught 70 percent of all real Dermatofibroma cases. In Experiment 3, the model only caught 40 percent.

An untrained observer might look at this drop in recall and declare the generative pipeline a failure. However, in machine learning, focusing on a single metric while ignoring the rest of the system leads to disastrous real-world consequences. To understand why Experiment 3 is actually the most successful and balanced model of the entire project, we have to analyze exactly why each metric shifted.

Why Recall Decreased (And Why That Is Good)

In Experiment 2.5, we used classical augmentation—rotating and flipping the same 71 images—and heavily penalized the model for missing the rare class. The model became highly sensitive to the heavy penalty. Because it lacked the visual diversity to understand what the class actually looked like, it adopted a skewed strategy. It began over-predicting "Dermatofibroma" anytime an image looked remotely ambiguous.

This over-prediction artificially inflated the recall to 0.7000. Yes, it caught 70 percent of the actual lesions, but it did so by throwing a massive, sloppy net over the entire dataset.

In Experiment 3, we injected 400 completely new, synthetic variations of the class. The neural network finally had enough synthetic visual variation to form improved decision boundaries. Because it possessed better examples to learn from, the network became more selective. It stopped over-predicting the class. It only diagnosed an image as a Dermatofibroma when the image actually possessed the correct structural characteristics observed during training.

Because the model became more careful and selective, the recall naturally dropped from the artificially inflated 0.7000 down to a realistic 0.4000. It missed a few more of the subtle real cases, but it did so because it was making measured, mathematical decisions rather than relying on a skewed strategy.

Why Precision Improved

The clearest evidence that the model became more selective is found in the precision metric.

Precision measures how often the model is correct when it decides to raise an alarm. In Experiment 2.5, the model raised so many false alarms that its precision dropped to 0.2692. Nearly three out of every four times the model diagnosed a Dermatofibroma, it was completely wrong. It was falsely labeling harmless moles as rare conditions.

In Experiment 3, the precision recovered to 0.4211. Because the synthetic images provided the morphological variation needed to better separate a harmless mole from a Dermatofibroma, the network made far fewer false accusations. It became a much more trustworthy diagnostic tool. If the model raised an alarm in Experiment 3, the doctor could have significantly more confidence that the alarm was genuine.

Why Overall Accuracy Recovered

In Experiment 2.5, the over-sensitive network was labeling thousands of ordinary, harmless moles (the Nevus class) as the rare class. Because ordinary moles make up the vast majority of the test set, getting them wrong destroyed the model's overall accuracy, dragging it down to 68.44 percent.

In Experiment 3, the synthetic data taught the network to be selective. Because it reduced its false positive rate on ordinary moles, its performance on the majority class skyrocketed back to normal. The overall accuracy of the system recovered to 0.7723.

This is an incredibly important achievement. The model achieved nearly the exact same overall accuracy as the original Baseline model (0.7763), but unlike the Baseline model, it was no longer completely blind to the rare class.

The Ultimate Judge: Macro F1-Score

The most important number in the entire results table is the Macro F1-Score.

As we established early in this project, we cannot judge an imbalanced medical system by accuracy alone, nor can we judge it by recall alone. The Macro F1-Score is the great equalizer. It calculates the balance between precision and recall for every single class independently, and then averages them together, treating the rarest class as mathematically equal to the most common class.

In the Baseline experiment, the Macro F1 was 0.5307. In the Classical Augmentation experiment, the Macro F1 was 0.5705. In the DCGAN Hybrid experiment, the Macro F1 reached 0.6166.

Experiment 3 achieved the highest Macro F1-Score of the entire project. This proves mathematically that the system reached its greatest state of balanced classification performance. The network learned to respect the minority class without cannibalizing the performance of the majority classes.

Analyzing the Confusion Matrix

To visually confirm this new state of balance, we can examine the final confusion matrix for Experiment 3.

[Placeholder: Figure showing the 7x7 Confusion Matrix for Experiment 3]

When we look at the Dermatofibroma column in the new matrix, the difference from Experiment 2.5 is stark. In Experiment 2.5, the column was smeared with numbers from top to bottom, representing the massive wave of false alarms.

In Experiment 3, the column has cleaned up significantly. The stray numbers scattering outside the correct diagonal line have dramatically reduced. The model is no longer wildly pulling images from the Benign Keratosis (bkl) and Nevus (nv) rows. The confusion matrix proves that the model successfully established improved decision boundaries between overlapping classes. The synthetic images successfully acted as a structural buffer, helping the network separate similar visual presentations.

The Central Conclusion of the Hybrid Experiment

This experiment fundamentally changes how we view success in imbalanced machine learning.

If a researcher's only goal is to push a single metric to its absolute maximum, they can simply use classical augmentation and heavy loss penalties to force the network to over-predict the minority class, artificially inflating recall. But this is not practical machine learning; it is mathematical brute force, and it creates models that are highly inaccurate on common cases.

The goal was never to maximize recall alone. The goal was to achieve balanced classification performance.

Experiment 3 demonstrates that synthetic visual variation generated by a DCGAN does not simply maximize minority recall. Instead, it serves to restore the balance between sensitivity and specificity. By providing the neural network with visually plausible, synthetic variations of a class, we give the network the information it needs to reduce over-prediction and establish improved decision boundaries. The Hybrid Pipeline allowed us to keep the system-wide accuracy high while maintaining a meaningful and precise detection capability for an extreme minority condition.

What Comes Next?

The classifier is now performing better, presenting observed downstream performance improvements. We have successfully used artificial intelligence to generate data, and we have proven mathematically that the generated data improves the system's overall balance.

However, we are still faced with one final, critical hurdle.

We know the model's metrics have improved, but we still do not know exactly where the model is looking when making its decisions. Neural networks are essentially black boxes. They output a diagnosis, but they do not provide a

visual explanation.

What if the model achieved a Macro F1-Score of 0.6166 by relying on background features? What if the synthetic images accidentally contained a strange background artifact, and the model simply learned to look for that artifact instead of looking at the actual skin lesions? In a high-stakes field like healthcare, numbers on a spreadsheet are never enough. A user must be able to verify what regions of the image influence the model's prediction.

To gather evidence that our model is acting on plausible visual features and not just hidden background artifacts, we must force the network to show us its internal attention. We must be able to point to the exact pixels on the photograph that triggered the prediction.

This requirement moves our project out of the realm of Data Augmentation and into the final frontier of the study: Explainable Artificial Intelligence. In our final chapter, we will deploy a visual mapping tool called Grad-CAM to look inside the final convolutional layers of our ResNet18 model and map its spatial attention.

Chapter Summary

In this chapter, we tested the true utility of our 400 synthetic Dermatofibroma images by injecting them into the training pipeline. We built a Hybrid Dataset containing 71 real images and 400 synthetic images, reducing the extreme class imbalance. To ensure a fair scientific control, we trained the exact same ResNet18 architecture and evaluated it on the exact same unseen Test Set.

The results proved that synthetic augmentation restores balanced classification performance. While classical augmentation (Experiment 2.5) artificially inflated minority recall to 0.7000 by causing the network to over-predict the rare class everywhere, the DCGAN Hybrid Pipeline (Experiment 3) provided enough synthetic visual variation for the network to establish improved decision boundaries. The model became more selective, causing recall to stabilize at 0.4000, while precision recovered from 0.2692 to 0.4211. Because the model reduced its false positive rate on harmless moles, overall accuracy recovered to 0.7723. Most importantly, Experiment 3 achieved a Macro F1-Score of 0.6166, the highest of the entire project, proving that generative augmentation supports a superior state of overall mathematical equilibrium. With the metrics validated, our final challenge is to look inside the black box and verify that the model is making its decisions by looking at the lesion itself rather than hidden background artifacts.

Chapter 9: Explainable Artificial Intelligence Using Grad-CAM

The model performs better, but can we trust where it is looking?

At the end of our previous chapter, we successfully trained our ResNet18 classifier on a Hybrid Dataset. By injecting 400 synthetic images of the rare Dermatofibroma class, we provided the network with the visual variation it needed to establish improved decision boundaries. The system stopped over-predicting the minority class, false alarms dropped, overall accuracy recovered, and the Macro F1-Score reached a project-high of 0.6166.

On paper, the experiment was a complete success. If we were working on a standard computer science problem, we might stop here. However, in the field of medical artificial intelligence, high mathematical metrics are not enough to deploy a system into the real world. A doctor cannot base a life-altering medical decision on a spreadsheet of numbers. They need to understand the evidence behind the decision.

In this chapter, we address the final and perhaps most critical challenge of our project: the black-box problem. We will explain the dangers of relying entirely on metrics, introduce the field of Explainable Artificial Intelligence (XAI), and detail how we used a visual mapping tool called Grad-CAM to peer inside the final layers of our neural network. By generating visual heatmaps, we will gather evidence to determine if our model is actually looking at skin lesions, or if it is simply cheating by looking at background artifacts.

The Black-Box Problem in Deep Learning

Deep neural networks are incredibly powerful, but they are fundamentally "black boxes." When you feed an image into a ResNet18 model, it passes through 18 layers of complex mathematical filters, activating millions of individual parameters. At the end of this process, the model simply outputs a prediction: "This is a Dermatofibroma with 92 percent confidence."

The model does not provide a written report. It does not circle the part of the image it found suspicious. It gives the human user absolutely no explanation for how it arrived at its conclusion.

In healthcare, this lack of transparency is incredibly dangerous. If a human doctor diagnoses a patient with cancer, the patient can ask, "Why do you think that?" The doctor will point to the specific irregular border of the mole, or note the asymmetrical color distribution. If an artificial intelligence makes

the same diagnosis, but cannot explain why, a doctor cannot blindly trust it.

The Danger of Shortcut Learning

Why is a black box so dangerous if its accuracy scores are high? The answer lies in a phenomenon known as "shortcut learning," or spurious correlation.

Neural networks are lazy. Their only goal is to minimize their mathematical error, and they will use any visual shortcut they can find to accomplish that goal. A famous example of this in artificial intelligence history involved a model trained to tell the difference between wolves and husky dogs. The model achieved near-perfect accuracy. But when researchers finally looked inside the black box, they discovered a catastrophic flaw. The model was not looking at the animals at all. It had simply noticed that all the husky photographs had snow in the background, while the wolf photographs had grass. The model had become a highly accurate "snow detector."

In medical imaging, shortcut learning can have fatal consequences. Often, when a doctor suspects a skin lesion is a dangerous melanoma, they will place a small physical ruler next to the lesion before taking the photograph, to document its size. Harmless moles rarely have rulers next to them. If you train a deep learning model on a dataset containing these images, the network might quickly realize that "ruler equals cancer."

If this happens, the model will achieve excellent accuracy and F1-scores during testing. But it has learned a spurious correlation. It is exhibiting the "Clever Hans" effect—appearing highly intelligent, but actually relying on an unintended background cue. If you take this model into a real hospital and show it a massive, deadly melanoma without a ruler in the background, it will confidently predict that the patient is perfectly healthy.

This is why performance metrics alone are insufficient. We must verify that the model's mathematical decision boundaries are anchored on the actual physical features of the skin lesion, and not on surgical rulers, background skin tones, or medical tape.

Introducing Explainable AI (XAI)

To solve the black-box problem, researchers developed a sub-field of machine learning known as Explainable Artificial Intelligence (XAI). The goal of XAI is to build tools that act as translators, converting the complex, multi-dimensional mathematics of a neural network into a format that humans can easily understand.

Over the past few years, several different XAI approaches have been developed:

LIME (Local Interpretable Model-agnostic Explanations): LIME works by breaking an image into tiny puzzle pieces, called super-pixels. It then slowly hides different puzzle pieces from the model and observes how the model's prediction changes. If hiding a specific puzzle piece causes the model's confidence to plummet, LIME assumes that piece was highly important. While useful, LIME's puzzle pieces do not always align with organic biological shapes, making it somewhat clunky for dermatology.

SHAP (SHapley Additive exPlanations): SHAP is based on game theory. It calculates the exact mathematical contribution of every single pixel toward the final prediction. In the early stages of this project, we actually attempted to use SHAP. However, we encountered severe technical limitations. SHAP is highly computationally expensive. Furthermore, when applied to dense, high-resolution color photographs of skin lesions, SHAP often produced heatmaps that looked like scattered, noisy static. The visual explanations were extremely difficult for a human to interpret.

Saliency Maps: Saliency maps calculate which individual pixels are the most sensitive. If changing a single pixel slightly causes a massive change in the prediction, that pixel lights up. Unfortunately, saliency maps often highlight random edges and background noise, failing to show the broader "big picture" of what the network is looking at.

Grad-CAM (Gradient-weighted Class Activation Mapping): Grad-CAM takes a completely different approach. Instead of looking at individual pixels, it looks at the final convolutional layers of the neural network. By analyzing the high-level mathematical filters right before the final prediction is made, it generates a smooth, flowing heatmap that highlights the broad spatial regions the model was paying attention to.

Why We Selected Grad-CAM

For this project, Grad-CAM was selected as the ideal XAI tool for several key reasons.

First, it is specifically designed to be compatible with Convolutional Neural Networks (CNNs) like our ResNet18 model. It requires no changes to the network's architecture and does not require us to retrain the model.

Second, the visual interpretability of Grad-CAM is excellent. Instead of outputting noisy static, it outputs smooth, color-coded heatmaps. This makes it incredibly easy for a human to look at an image and immediately understand the

model's broad spatial attention.

Finally, it is computationally practical. Unlike SHAP, which requires massive amounts of processing power to test thousands of pixel combinations, Grad-CAM requires only a single forward and backward pass through the network. This allowed us to quickly generate heatmaps for any image in our Test Set.

Understanding Grad-CAM Intuitively

To understand how Grad-CAM works without relying on heavy calculus, we can use a simple analogy.

Imagine the final convolutional layer of our ResNet18 model is a committee of 512 different detectives looking at a photograph of a skin lesion. Every detective is searching for a different visual clue. Detective 1 is looking for jagged borders. Detective 2 is looking for dark brown pigments. Detective 3 is looking for circular shapes.

When the photograph is shown to the committee, each detective creates a map of where they found their specific clue. These maps are the "feature maps" of the neural network.

Now, the committee has to make a final diagnosis. To do this, a supervisor (the final mathematical gradient) steps in. The supervisor knows which clues are actually important for diagnosing a specific disease. If the goal is to diagnose a Dermatofibroma, the supervisor might say, "Detective 1 (jagged borders), your clue is not important for this disease; I am going to silence your map. Detective 3 (circular shapes), your clue is highly important; I am going to amplify your map."

Grad-CAM simply takes the feature maps from all 512 detectives, weights them based on how loudly the supervisor amplified them, and stacks them all on top of each other.

The result is a single, unified heatmap. The areas where the most important detectives found their clues glow bright red (high attention). The areas where no important clues were found fade to dark blue (low attention).

By overlaying this heatmap directly on top of the original photograph, we can see exactly which spatial regions drove the final prediction.

Applying Grad-CAM to the Hybrid ResNet18 Model

To execute this analysis, we loaded our final, locked ResNet18 model from

Experiment 3.

We specifically targeted the very last convolutional block in the ResNet architecture, technically known as layer4[-1]. We targeted this specific layer because it represents the deepest level of spatial understanding in the network. If we targeted an early layer, the heatmap would simply highlight basic edges and lines. By targeting the final layer, the heatmap highlights the complex, high-level morphological structures the network learned right before it handed the information off to the final classification head.

We then took a random batch of images from our completely unseen Test Set. We fed these images into the network and asked the network to predict the disease.

To create a clear visualization, we had to reverse some of our earlier data processing. During training, we had normalized the pixel colors of the images to help the network learn mathematically. To show the images to a human reader, we had to mathematically "un-normalize" the pixels, restoring the photographs to their original, natural colors. Finally, we generated the Grad-CAM heatmaps for the model's predicted class, and projected the color-coded maps over the natural photographs.

Presenting the Grad-CAM Results

When we generated the visualizations and examined the heatmaps, the results provided highly encouraging evidence regarding the model's behavior.

[Placeholder: Figure 9.1 - Grid showing original images on the left, and Grad-CAM heatmaps overlaid on the right. Red areas indicate high model attention, while blue areas indicate low attention.]

When reviewing the visual outputs across various classes, we observed several consistent patterns:

1. **Attention Concentrated on Lesion Regions:** In the vast majority of the test images, the bright red and yellow "hot zones" of the heatmap were anchored squarely over the physical skin lesions. The model was not looking randomly at the corners of the photograph. Its mathematical attention was focused almost exclusively on the area where the biological abnormality was located.
2. **Limited Attention to Background Artifacts:** Crucially, we observed what we term "Zero Clever Hans" behavior. Many images in the HAM10000 dataset contain thick, dark body hair crisscrossing the skin. Other images feature bright lighting glares or shadows. In our visualizations, the Grad-CAM heatmaps consistently ignored these features. The body hair and background skin tones

remained in the dark blue, "cold" zones of the heatmap. This provides strong evidence that the network learned to bypass spurious correlations and focus on the lesion itself.

3. Focus on Borders and Internal Cores: When inspecting the hot zones more closely, we noticed that the model's attention often traced the outer, irregular boundaries of the lesions, or pooled densely in the darkly pigmented central cores. This aligns with the visual variations we attempted to provide via our DCGAN in the Hybrid pipeline. The synthetic data provided varied borders and shapes, and the classifier learned to seek out those specific spatial transitions.

Interpreting the Heatmaps

The most fascinating insight came from analyzing the model's mistakes.

In one specific visualization, the model was shown an image of a Benign Keratosis (bkl), which is a harmless age spot. However, the model incorrectly predicted that the image was a Melanoma (mel), the deadliest form of skin cancer.

If we only looked at the performance metrics on a spreadsheet, we would simply log this as a failure—a false positive that lowers our precision. But when we looked at the Grad-CAM heatmap for this specific mistake, the visual explanation revealed something profound.

In the clinical world, Benign Keratosis is notoriously difficult to diagnose because it often presents with dark, asymmetrical borders. It is widely known among dermatologists as the "Great Mimic" because it looks terrifyingly similar to Melanoma.

When we examined the Grad-CAM heatmap for this incorrect prediction, the red hot-zone was heavily concentrated entirely on a dark, highly asymmetrical, jagged edge of the harmless lesion.

The network made a mathematical mistake, but it was a "smart" mistake. It did not guess Melanoma because it saw a surgical ruler, or because of a random shadow. It guessed Melanoma because it looked directly at a dark, jagged border—a classic visual warning sign of cancer—and made a cautious, health-first decision.

This type of explanation is exactly why XAI is required in healthcare. Even when the model is wrong, understanding why it is wrong allows a human doctor to trust the system's underlying logic. A doctor looking at this heatmap would

immediately understand why the AI raised an alarm, and could then use their superior clinical experience to overrule it safely.

Limitations of Grad-CAM

While these visualizations are highly encouraging, we must be scientifically honest about their limitations. It is very easy to look at a Grad-CAM heatmap highlighting a tumor and excitedly claim, "The model reasons exactly like a doctor!"

This is a dangerous overstatement. Grad-CAM provides evidence of spatial attention, not proof of medical reasoning.

When a red hotspot appears over a jagged border, Grad-CAM tells us where the model is looking. It does not tell us what the model is thinking. We do not know if the model is looking at the border because it understands the biological concept of "asymmetry," or simply because it detects a sharp contrast between dark and light pixels.

Furthermore, Grad-CAM heatmaps are inherently low-resolution. Because they are generated from the final, highly compressed convolutional layer of the network, the heatmaps appear as large, cloudy blobs rather than pixel-perfect outlines. They cannot explain highly microscopic features, such as individual cellular structures or tiny, sub-millimeter pigment dots.

Grad-CAM is an approximate explanation. It is a supporting piece of evidence, not absolute proof. However, as a diagnostic tool for finding catastrophic failures—like a model anchoring on a surgical ruler—it is highly effective.

Conclusion of the Experimental Journey

With the completion of the Grad-CAM analysis, we have reached the end of our experimental journey.

We started by identifying a severe class imbalance problem in the HAM10000 dataset. We proved that standard accuracy is an illusion, and we showed how mathematical penalties (Weighted Loss) and Classical Augmentation cause a network to adopt a skewed, over-predicting strategy.

To solve this, we introduced Generative Artificial Intelligence. We stabilized a DCGAN on a tiny micro-dataset to hallucinate synthetic visual diversity. By injecting these generated images into a Hybrid pipeline, we restored the model's mathematical equilibrium, achieving our highest balanced performance. Finally, through Explainable AI, we validated that the model achieved these metrics by

anchoring its attention on plausible morphological boundaries, rather than cheating by using background artifacts.

The final model has now been evaluated both quantitatively (through rigorous metrics) and qualitatively (through visual explanations).

However, no scientific study is perfect. Throughout this project, we made specific engineering choices, accepted certain hardware constraints, and left several questions unanswered. To complete this research paper, we must turn a critical eye on our own work. In the final chapter, we will discuss the limitations of our methodology, the remaining challenges, and the future directions this research must take before such a system could ever be deployed in a real-world clinical setting.

Chapter Summary

In this chapter, we addressed the "black-box" problem of deep learning. High performance metrics are insufficient in healthcare because models can easily achieve high scores by learning spurious correlations, such as relying on background surgical rulers rather than medical pathology (the Clever Hans effect). To verify our Hybrid ResNet18 model's behavior, we utilized Explainable AI, specifically targeting the final convolutional layer with Grad-CAM to generate visual heatmaps. The results demonstrated that the model anchored its mathematical attention squarely on the physical boundaries and pigmented cores of the lesions, successfully ignoring background noise, skin tones, and body hair. Furthermore, analyzing misclassifications revealed that the model's mistakes were grounded in plausible visual overlap, such as confusing the asymmetrical borders of a harmless age spot for melanoma. While we carefully note that Grad-CAM proves spatial attention rather than actual clinical reasoning, the heatmaps provide strong supporting evidence that the generative pipeline helped the model establish decision boundaries based on relevant morphological variation. With the system now validated both quantitatively and qualitatively, we move to our final discussion of project limitations and future research directions.

Chapter 10: Limitations, Challenges, and Future Directions

No scientific study is perfect. In the world of academic research, discussing limitations is just as important as celebrating successes. When a researcher hides the flaws in their work, they create a false sense of security. This is especially dangerous in medical artificial intelligence, where overconfidence

can lead to physical harm. By speaking honestly about our limitations, we do not invalidate our results. Instead, we define the strict boundaries of what conclusions can be drawn, and we leave a clear map for future researchers to follow.

In this chapter, we will take a highly critical look at our own project. We will discuss the practical hurdles that shaped our engineering decisions, the statistical limitations of our experiments, the boundaries of our generative model, and the reasons why this system is not yet ready for a real hospital. Finally, we will reflect on the most important lessons learned throughout this intellectual journey, and explore how future research can build upon this foundation.

Section 10.1 – Challenges Faced During the Project

The design of this project was entirely shaped by the real-world obstacles we encountered. Every architectural choice and training strategy was a direct response to four major challenges.

The first challenge was severe class imbalance. We set out to build a diagnostic tool, but the HAM10000 dataset handed us a massive mathematical hurdle. The dataset contained 4,683 images of ordinary moles, but only 71 training images of the rare Dermatofibroma class. This extreme 70:1 imbalance ratio forced us to completely abandon standard training methods. We could not simply feed the data into a network; we had to actively intervene using weighted penalties and synthetic data generation.

The second challenge was the underlying reality of data scarcity. When a neural network fails to learn a concept, the standard industry advice is simply to "collect more data." In the medical domain, this is incredibly difficult. Obtaining medical images involves navigating strict patient privacy laws. Furthermore, an image is useless to a supervised learning algorithm unless it is accurately labeled, which requires hours of expensive labor from highly trained dermatologists. Because the Dermatofibroma condition is relatively rare, gathering hundreds of new, real photographs was impossible for this study. This scarcity was the direct reason we turned to generative artificial intelligence.

The third challenge was the sheer instability of Generative Adversarial Networks. Training a GAN is notoriously difficult even under the best conditions. When we attempted to train a Discriminator and a Generator on a micro-dataset of only 71 images, the system immediately broke down. We faced Discriminator domination, where the detective network perfectly memorized the training images and gave the Generator zero helpful feedback. We also faced mode collapse, where the Generator produced chaotic static instead of diverse

morphological variations. Overcoming this instability required us to heavily modify the training loop with techniques like Discriminator Dropout and Asymmetric Update Ratios.

The fourth challenge was computational constraints. This project was executed in a Kaggle notebook environment. While cloud-based environments democratize access to machine learning, they come with strict limitations on memory and graphics processing unit (GPU) time. We did not have a massive server farm at our disposal. This reality forced us to be highly practical. We could not use massive, 152-layer neural networks, nor could we use incredibly heavy generative models like Diffusion Models. We specifically selected ResNet18 and DCGAN because they were mathematically efficient and compatible with our limited hardware, forcing us to rely on smart engineering rather than brute computational force.

Section 10.2 – Experimental Limitations

Because we operated under these constraints, our experimental design carries specific scientific limitations that must be acknowledged.

Single Dataset Limitation: This study was conducted entirely on the HAM10000 dataset. While HAM10000 is a large and respected dataset, it was collected from specific clinical centers and primarily features patients with lighter skin tones. Neural networks are highly sensitive to the data they are trained on. Because we did not train our model on diverse global populations, we cannot guarantee that the model would perform equally well on patients with darker skin tones. The results may not automatically generalize to all demographic groups.

Single Seed Study: Neural networks contain an element of randomness. When you first create a model, its internal mathematical weights are initialized randomly using a number called a "seed." For this project, all experiments were conducted using a single random seed (seed=42). If we changed the seed, the starting weights would change, and the final accuracy metrics might shift slightly up or down. Because we only ran the experiment once, no multi-seed statistical analysis was performed. We cannot definitively calculate a mathematical margin of error or prove strict statistical significance. We can only report the observed downstream performance improvements.

No Cross-Validation: In this study, we separated our data into one strict Training Split, one Validation Split, and one Test Split. While this is standard practice, a more rigorous approach is K-Fold Cross-Validation. In K-Fold validation, the dataset is split and tested multiple times, with different groups of images taking turns as the Test Set. This ensures the model is evaluated on every single image in the dataset. Because training multiple

generative pipelines is computationally expensive, K-Fold validation was not performed.

No External Validation Dataset: Our Test Set was strictly isolated from our Training Set, but all the images still came from the HAM10000 collection. In the real world, different hospitals use different cameras, different lighting, and different dermatoscope fluids. An artificial intelligence model often experiences a drop in performance when it is moved from its home hospital to a completely new, external clinic. Our model was not tested on completely independent, external datasets, meaning its true real-world robustness remains unverified.

Small Minority Class Limitation: Finally, we must acknowledge the absolute mathematical ceiling of our data. We generated 400 synthetic images, but those synthetic images were ultimately derived from the patterns hidden inside only 71 real photographs. There is a limit to how much diversity a generative model can extract from 71 examples. This tiny foundation fundamentally limits what any downstream classification model can ultimately learn.

Section 10.3 – Generative Model Limitations

We must also be incredibly clear about the limitations of our DCGAN approach. Generative models are powerful, but they are not magic.

In most modern computer vision research, the quality of a generated image is evaluated using complex mathematical formulas like the Fréchet Inception Distance (FID) or the Inception Score. These formulas compare the statistical distributions of the real and fake images. In this project, no FID score and no Inception Score were calculated. Attempting to run these statistical formulas on a dataset of only 71 images produces noisy, meaningless math.

Because we could not rely on mathematical formulas, our synthetic image quality was evaluated purely based on two functional criteria: visual inspection to ensure the model was not producing chaotic static, and downstream classifier performance. If the synthetic images improved the Macro F1-Score of the classification model, we deemed them useful.

Furthermore, there was no expert dermatologist review of the generated images. We do not know if the synthetic images look clinically plausible to a trained medical professional. Therefore, the generated images should not be interpreted as clinically validated lesions. They are simply synthetic visual variations—mathematical buffers that successfully helped a neural network draw better decision boundaries. We make no claims regarding their absolute biological realism.

Section 10.4 – Explainability Limitations

In the final phase of our project, we used Explainable Artificial Intelligence (XAI) to ensure the model was not relying on background artifacts. We generated visual heatmaps using Grad-CAM. While these heatmaps provided highly encouraging evidence, they come with significant limitations.

Grad-CAM provides attention maps. It shows us where the model is looking, but it does not reveal the network's full reasoning. If a heatmap highlights the jagged edge of a skin lesion, we know the border influenced the prediction. However, we do not know if the network understands the medical concept of "asymmetry," or if it simply likes the sharp contrast of dark and light pixels.

Furthermore, Grad-CAM heatmaps are approximate explanations. Because they are generated from the highly compressed final layers of the network, they appear as broad, cloudy zones of color. They cannot highlight microscopic details, like specific cell patterns or tiny pigment networks.

Grad-CAM cannot prove clinical reasoning. It is a fantastic tool for debugging a model and ensuring it is not cheating by looking at a surgical ruler, but it is only supporting evidence. Truly understanding the complex internal thoughts of a deep neural network remains a massive, open research challenge in the field of artificial intelligence.

Section 10.5 – Clinical Deployment Limitations

Given the promising metrics achieved by the Hybrid pipeline, it is tempting to ask: "When can we put this in a hospital?" The honest scientific answer is that the current system is absolutely not ready for clinical deployment. This project is a research prototype, not a clinical diagnostic device.

Before an artificial intelligence system can be allowed to influence real patient care, it must pass through rigorous, multi-stage validation. First, the model would require extensive dermatologist validation, where human experts manually verify the model's predictions and interpretability maps. Second, it would require strict bias and fairness testing to ensure it does not perform worse on underrepresented demographics or specific skin tones.

Third, the system would need to be tested in multi-center clinical trials. It would have to be deployed silently in multiple hospitals around the world, analyzing images from completely different camera systems to ensure it does not suffer from domain shift. Finally, the system would have to navigate complex regulatory approval requirements, such as those governed by the Food and Drug

Administration. The safety of the patient must always outweigh the excitement of high mathematical metrics.

Section 10.6 – Future Research Directions

The limitations of our study naturally outline the path for future research. If this project were to be extended, several exciting directions could be explored.

The most obvious improvement would be training the system on larger, multi-dataset collections. By combining HAM10000 with other international databases, researchers could expose the neural network to a much wider variety of skin tones, camera types, and rare disease presentations, greatly improving real-world robustness.

Future researchers could also explore advanced generative models. With more computational power, architectures like StyleGAN or modern Diffusion Models could be applied to medical image generation. If the initial dataset size is large enough, researchers could calculate formal quantitative GAN evaluations, such as the Fréchet Inception Distance, to mathematically verify the quality of the generated medical images.

To improve the statistical rigor of the classification metrics, future studies must implement multi-seed statistical analysis. By running the experiments multiple times with different random initializations and averaging the results, researchers can provide strict confidence intervals to prove that the performance gains are mathematically significant.

Explainable AI can also be vastly improved. Researchers could combine Grad-CAM with other, more precise interpretability methods like SHAP or Integrated Gradients. By using multiple XAI methods simultaneously, researchers can provide doctors with much sharper, pixel-level explanations of the network's behavior.

We can also look beyond the ResNet18 architecture. In recent years, Vision Transformers have emerged as a powerful alternative to standard Convolutional Neural Networks. Researchers could explore Hybrid CNN-Transformer systems, which combine the local texture-finding abilities of a CNN with the global, big-picture awareness of a Transformer.

Finally, future systems must become multi-modal. When a human dermatologist diagnoses a patient, they do not just look at a photograph. They look at the patient's age, their biological sex, the location of the lesion on the body, and the patient's medical history. Our current model only looks at pixels. A truly robust clinical artificial intelligence system must be multi-modal, combining image analysis with patient metadata and clinical notes to make a holistic,

informed decision.

Section 10.7 – Lessons Learned

This project was a deep intellectual journey. We began with a highly imbalanced dataset and ended with a balanced, explainable diagnostic prototype. Along the way, we learned several critical lessons about the reality of applied machine learning.

We learned that accuracy can be highly misleading. When a dataset is imbalanced, a model can achieve exceptional accuracy by simply taking the lazy route and guessing the majority class every time. In healthcare, missing a rare disease is unacceptable, which is why balanced metrics like the Macro F1-Score matter far more than raw overall accuracy.

We learned that data leakage can invalidate an entire experiment. Because dermatologists take multiple photographs of the exact same physical skin bump, random data splitting causes the network to memorize patients rather than learn diseases. By building a strict, lesion-level stratified split, we created a harsh but honest experimental environment.

We learned that more images do not necessarily mean more information. When we used classical augmentation to rotate and zoom our images, the network saw thousands of new pixel grids, but no new morphological variations. This caused the network to become terrified of missing the rare class, resulting in a massive wave of false alarms.

We learned that GAN training on tiny datasets is incredibly difficult. Extracting synthetic visual variation from only 71 images requires careful, deliberate engineering to prevent the Discriminator from dominating the game.

Finally, we learned that Explainability is absolutely essential in healthcare artificial intelligence. High metrics on a spreadsheet are meaningless if the model is cheating by looking at a surgical ruler in the background. Tools like Grad-CAM are required to bridge the gap between machine learning mathematics and human clinical trust.

Although important limitations remain regarding statistical sample sizes and clinical validation, the project demonstrated a clear and powerful concept. By combining careful data engineering to prevent leakage, generative artificial intelligence to provide synthetic visual variation, and explainable tools to verify model attention, we can substantially improve the balanced detection of rare classes in highly imbalanced medical datasets. With this final reflection complete, we turn to our overall conclusion to summarize the impact of this

study.

Chapter Summary

In this chapter, we conducted a critical review of our own methodology. We discussed the practical challenges that shaped the project, including the extreme 70:1 class imbalance, data scarcity, the mathematical instability of training a GAN on only 71 images, and the computational constraints of a cloud-based notebook environment. We honestly outlined our experimental limitations, noting that this was a single-seed study without cross-validation, evaluated on a single dataset. We acknowledged that our synthetic images were evaluated functionally rather than via mathematical formulas like FID, and that our Grad-CAM heatmaps provide approximate spatial attention rather than proof of clinical reasoning. We emphasized that this system is a research prototype, completely unready for hospital deployment without massive regulatory and multi-center validation. These limitations pave the way for future research, including the use of Vision Transformers, multi-modal systems, and multi-seed statistical trials. Ultimately, the lessons learned—that accuracy is an illusion, volume does not equal diversity, and balance is more important than recall—form the intellectual core of the project. Although limitations exist, the study successfully proves that generative augmentation and explainable AI can work together to restore equilibrium to imbalanced medical classifiers.

Chapter 11: Conclusion

The journey of this project began with a simple but profound medical reality. Skin cancer is one of the most common diseases in the world, and early diagnosis is the ultimate key to patient survival. While artificial intelligence presents an incredible opportunity to assist doctors by serving as a tireless second opinion, bridging the gap between computer science and real-world clinical application is vastly more complicated than simply feeding photographs into a neural network. Real-world medical data is messy, constrained, and inherently biased. This project was not just an exercise in writing code; it was a deeply educational exploration into why deep learning models fail on medical data, and how we can use careful engineering, generative artificial intelligence, and visual interpretability to guide them back on track.

The foundational challenge of this entire study was the problem of extreme class imbalance. In the real world, harmless moles are common, while specific diseases like Dermatofibroma are rare. When we examined the HAM10000 dataset, we found

this reality reflected in the numbers: there were thousands of images of ordinary moles, but only 71 training images of the rare Dermatofibroma class, creating a staggering 70:1 imbalance ratio. Deep learning models are inherently lazy. They seek to minimize their mathematical error using the easiest path available.

Before we even attempted to train a model to solve this imbalance, we had to ensure our experimental environment was perfectly honest. We discovered that standard data splitting methods create catastrophic data leakage in medical imaging, because doctors often take multiple photographs of the exact same physical skin lesion. If we allowed photos of the same physical bump to appear in both the training set and the test set, the neural network would simply memorize the patient rather than learn the disease. By implementing a strict, lesion-level stratified split, we built a mathematically impenetrable wall between our training and testing data. This crucial first step guaranteed that every success and every failure we recorded later in the project was genuine.

With a trustworthy environment established, our experimental journey revealed a series of profound lessons about machine learning metrics. When we trained our standard Baseline ResNet18 model on the raw, imbalanced data, the results were highly deceptive. The model achieved an impressive overall accuracy of over 77 percent. To an untrained observer, the system appeared to be a massive success. However, deeper analysis revealed that the model had completely ignored the rare Dermatofibroma class, scoring zero in precision and zero in recall. Because the dataset was flooded with ordinary moles, the network realized it could achieve a high accuracy score simply by guessing the majority class every single time. This baseline experiment proved conclusively that in the realm of medical artificial intelligence, overall accuracy is a dangerous illusion.

To correct this blind spot, we engaged in a step-by-step scientific ablation study. We first attempted to fix the problem using mathematics alone. By applying Weighted Loss, we made mistakes on the rare disease incredibly expensive. The model woke up and paid attention, but we quickly realized that changing the grading rubric does not magically provide the network with new visual knowledge.

We then attempted to solve the knowledge gap by using classical data augmentation. We rotated, flipped, and zoomed the 71 rare images to artificially expand the dataset. This led to our most educational failure. Because the network was terrified of the heavy mathematical penalties but lacked true morphological variation to learn from, it adopted a panicked strategy. It began guessing the rare disease anytime an image looked remotely ambiguous. While this artificially pushed the minority recall up to 70 percent, the precision collapsed to roughly 26 percent, and the global accuracy dragged downward. We

learned a critical rule of computer vision: generating more volume from existing images does not equal generating new visual diversity. A rotated image of a tumor is still the exact same tumor.

To break this deadlock, we turned to Generative Artificial Intelligence. We hypothesized that if we could hallucinate entirely new, visually plausible examples of the rare disease, we could provide the network with the structural variation it desperately needed. Training a Deep Convolutional Generative Adversarial Network (DCGAN) on a micro-dataset of only 71 images pushed the limits of mathematical stability. By engineering strict interventions—slowing the discriminator down, applying dropout to induce mathematical amnesia, and using asymmetric training loops—we stabilized the adversarial game and successfully generated 400 new synthetic images.

When we injected these 400 synthetic images into the training pipeline, we achieved the definitive breakthrough of the project. The DCGAN Hybrid Pipeline reached an overall accuracy of 0.7723, successfully recovering from the collapse caused by classical augmentation. More importantly, the system achieved a Macro F1-Score of 0.6166, the highest performance of the entire study.

The Macro F1-Score is the absolute most meaningful result we recorded. It is the great equalizer of machine learning metrics, treating the rarest disease as mathematically equal to the most common one. By achieving a project-high Macro F1, we proved that the synthetic visual variation allowed the network to stop wildly over-predicting the rare class and start drawing improved, cautious decision boundaries. The goal of this project was never to push minority recall to its absolute mathematical maximum at the expense of the rest of the system. The goal was to build a balanced, trustworthy classifier that respects rare conditions without destroying its ability to recognize common ones. The Hybrid Pipeline successfully restored that delicate mathematical equilibrium.

However, high metrics on a spreadsheet do not generate clinical trust. The final phase of our journey involved confronting the "black box" nature of deep learning. We had to prove that our balanced classifier was actually looking at skin lesions, rather than cheating by finding hidden background artifacts like surgical rulers or body hair. By deploying Explainable Artificial Intelligence, specifically Grad-CAM, we peered into the final spatial layers of the ResNet18 model. The resulting visual heatmaps provided highly encouraging supporting evidence. The network anchored its mathematical attention squarely on the physical boundaries and pigmented cores of the skin lesions. Even when the model made mistakes, the heatmaps revealed that those mistakes were grounded in plausible visual overlap, such as confusing the jagged, asymmetrical border of a benign age spot for a dangerous melanoma. Explainability transformed our model from a blind number-generator into a transparent system whose behavior could be

audited and understood.

Ultimately, the most valuable outcome of this research is not the final accuracy percentage, but the intellectual framework established along the way. We learned that an effective medical artificial intelligence system requires a harmony of disciplines. It requires strict data engineering to prevent leakage and guarantee honest evaluation. It requires an understanding of statistical metrics to see past the illusion of accuracy. It requires the creative application of generative models to overcome the physical limitations of real-world data collection. And finally, it requires the humility of explainable tools to verify that our artificial systems are making decisions for the right reasons.

As we look toward the future of AI-assisted medical diagnosis, the road ahead is long but incredibly promising. The systems of tomorrow will not replace human doctors; they will serve as highly advanced collaborative tools. To reach that future, the research community must continue to build upon these foundations. Future systems will need to move beyond single datasets, training on massive, globally diverse populations to ensure fairness across all skin tones. They will need to evolve from analyzing single images to becoming multi-modal systems that can read patient histories and genetic markers. They will require strict, multi-center clinical trials and deep regulatory scrutiny before they ever touch a patient's life.

Yet, despite the immense challenges that remain, studies like this one provide a clear and optimistic proof of concept. We have demonstrated that even when faced with extreme data scarcity and catastrophic class imbalance, careful, honest, and creative engineering can guide a neural network back to a state of balance. By continuing to prioritize system-wide equilibrium and absolute transparency over superficial metrics, the artificial intelligence community will steadily move closer to building diagnostic tools that are not only mathematically powerful, but clinically trustworthy.